
MySQL 8.0 接続設定ガイド

Connection Configuration Guide



CONTENTS

P.3	What is a database?
P.11	RDB server setup
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P.71	Communication settings

データベースとは

A database is a system that stores and manages data using application software. It not only records data, but also searches and organizes data at high speed and in a wide range of ways.

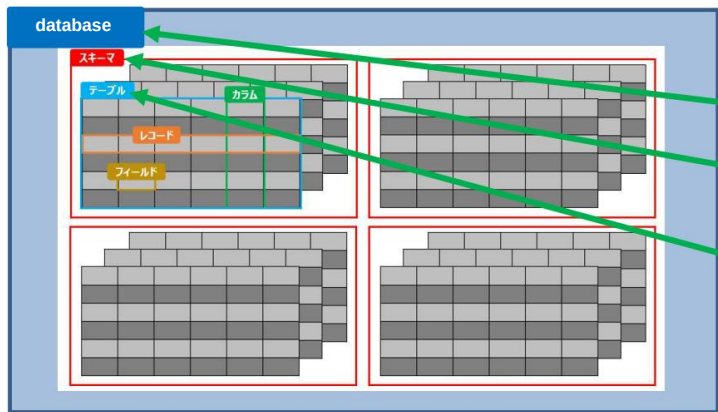
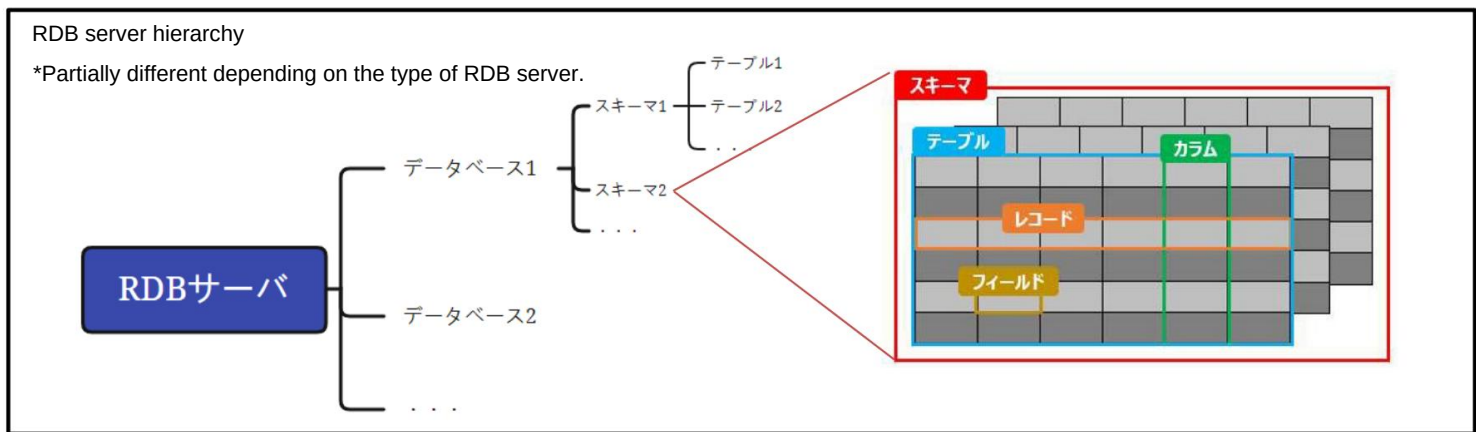
Compared to the CSV format data management method, the following mechanism enables high-speed processing. **The most important feature is the ability to format and record data in advance.**

database			
CSV			

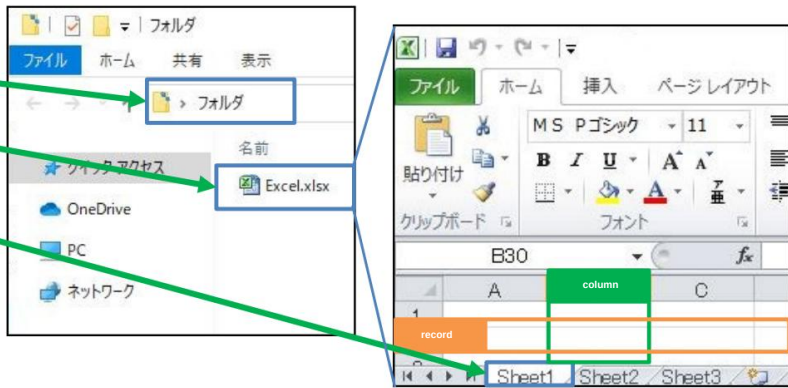
データベース用語について

Generally speaking, database refers to **RDB (relational database)**.

Like Excel, **RDB** manages data in tabular format and can have multiple tables like Excel sheets. Each table consists of columns and rows (records). Define the data type and rules of the data to be registered in the column, and the column name in advance. Data is added as a set of multiple registered columns, and the number of records (rows) increases.



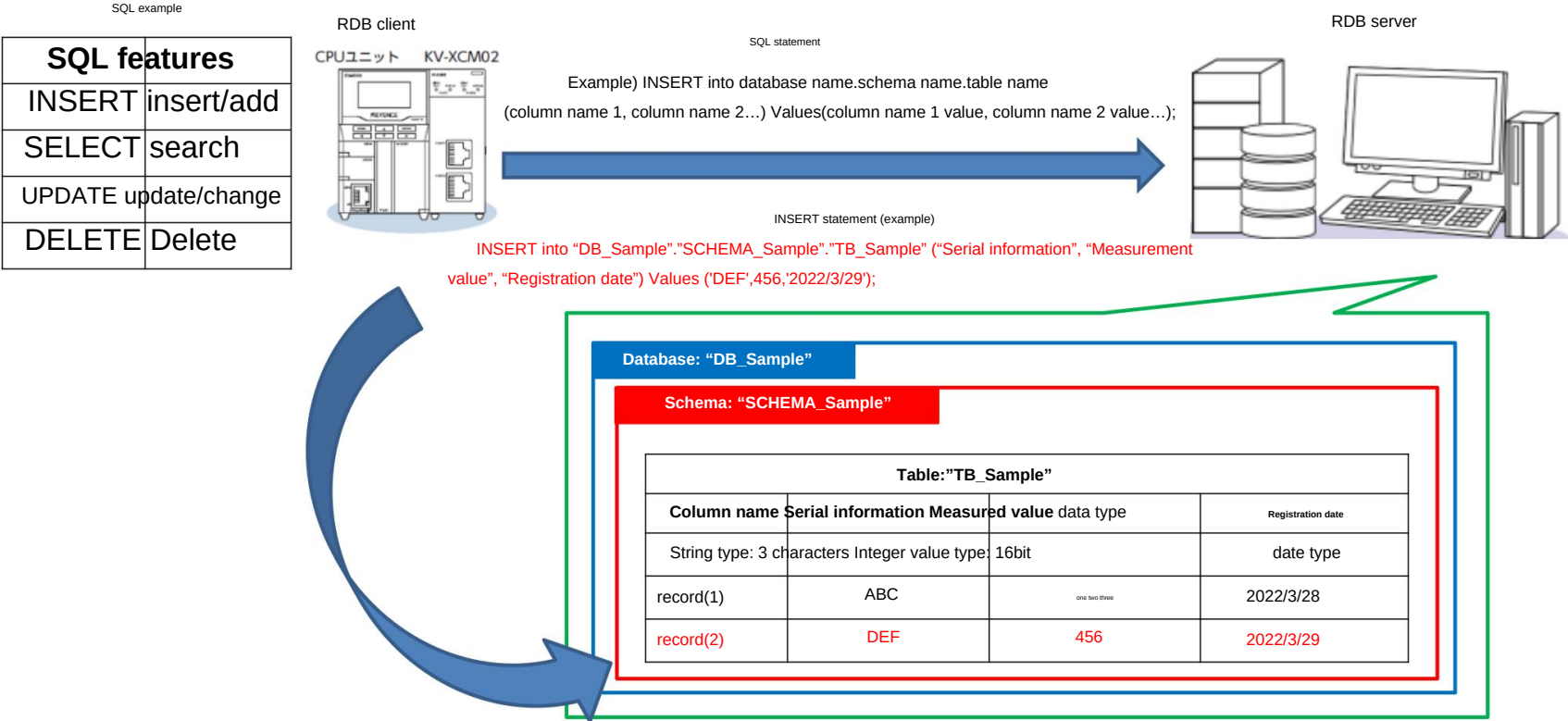
Comparison image with CSV (Excel)



データベースのデータ操作について

In RDB, data operations (additions, searches, changes, deletions, etc.) are performed on predefined table structures.
 Uses a language exclusively for RDB called SQL. Data operations are performed by sending and receiving syntax written in SQL (SQL statements).
 Here, the device that receives SQL statements is called the RDB server, and the KV (external device) that sends the SQL statements is called the RDB client.

The syntax of SQL statements is basically the same, but some parts may differ depending on the RDB server.



Data operations can be performed by specifying column names and column values, making searches, changes, and deletions fast and easy.

*For details on SQL statements, please check the RDB server manual.

データベースとの接続について

Since the RDB server is built on a PC, a connection to the PC is required for the RDB client to send SQL statements. If anyone can connect and view/edit data and perform operations on the PC, it could cause trouble. Therefore, as a security measure, user management is required on the RDB server as well as accounts for operating PCs. *For RDB server user management, there are different types of password authentication methods such as fingerprint authentication and face authentication.

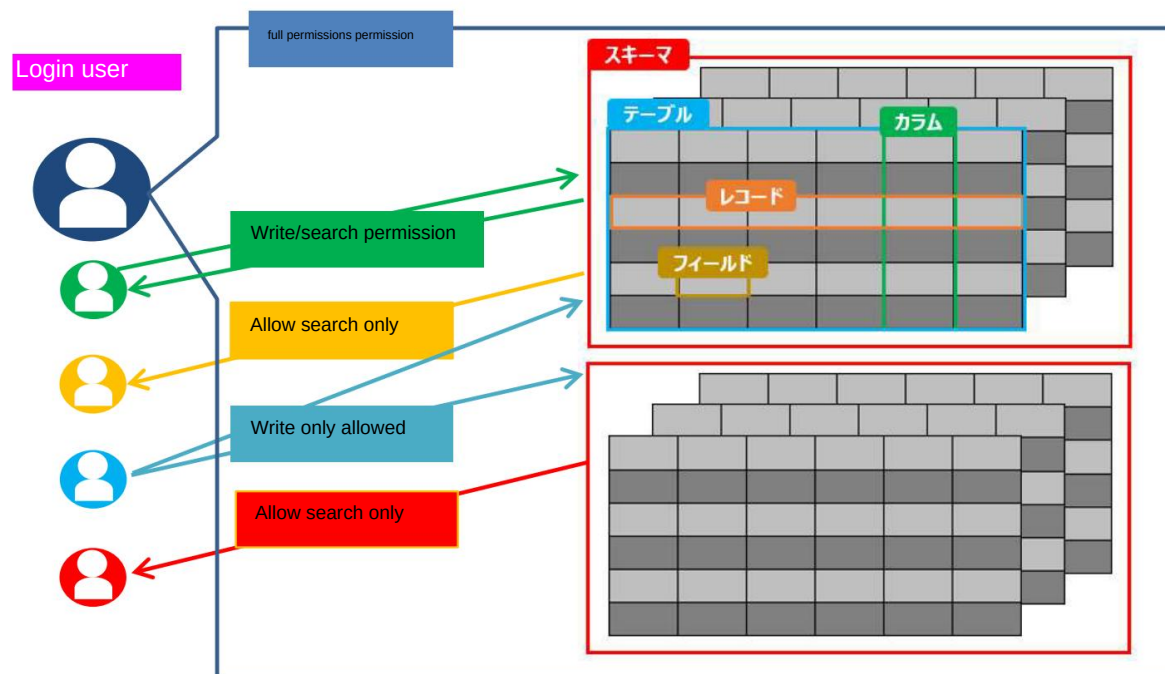
The RDB client uses the login user to connect to the RDB server. It is also necessary

to set various permissions for each logged-in user. •**RDB**

server management privileges : Login privileges, table creation privileges,

etc. •**Table operation privileges** : Data search privileges, data write

privileges, etc. For details regarding privileges, please check the RDB server manual.



データベースとの接続の流れ

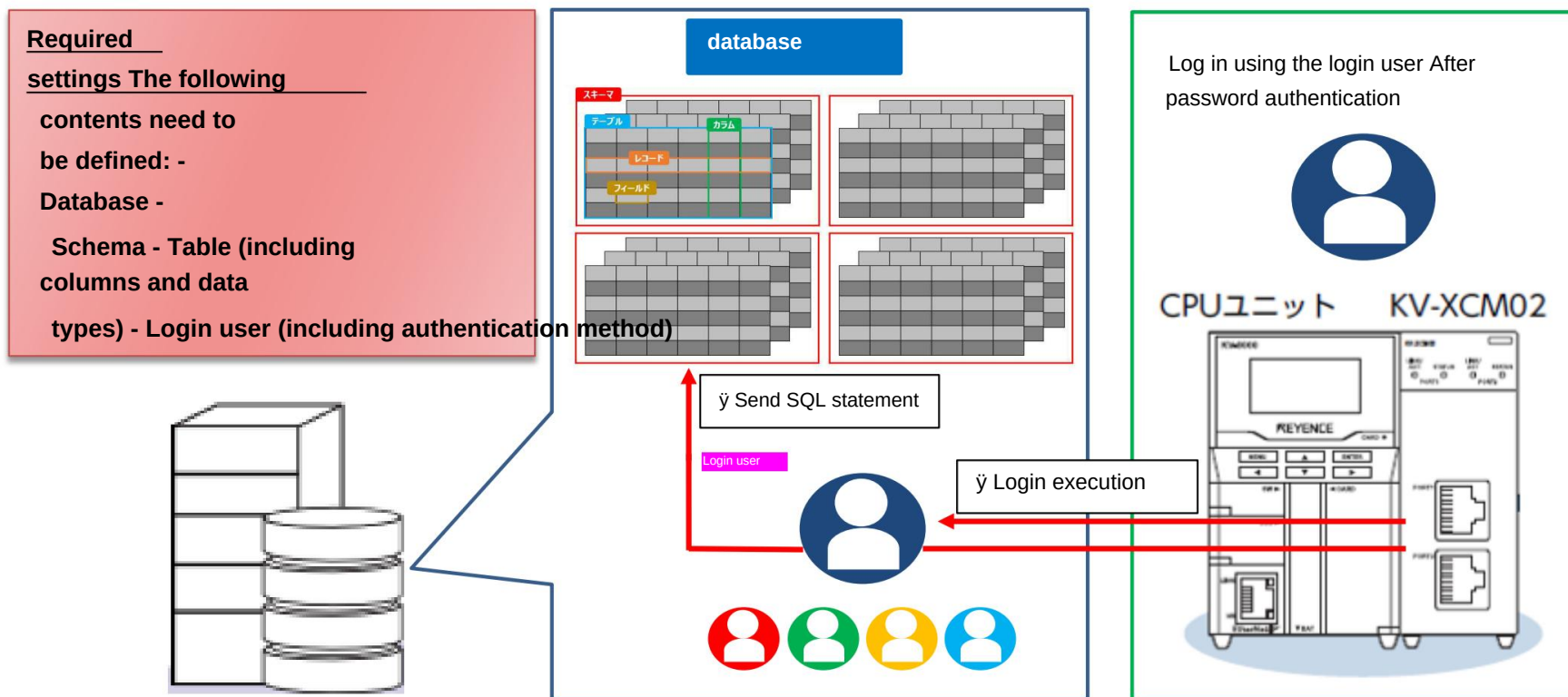
This section explains the flow of connection with an RDB client.

ÿ Connect to the RDB server by specifying the database name, login user name, and password.

*The RDB server password authentication method must be set to an authentication method supported by KV-XCM02.

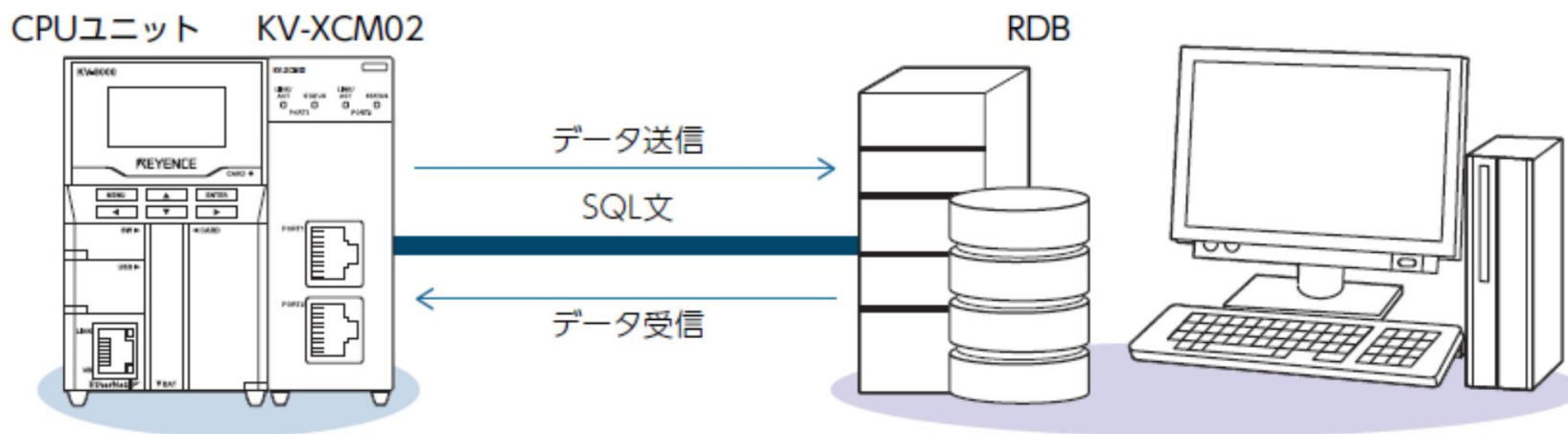
ÿ Manipulate data by sending SQL statements (database name, schema name, table name, and column name must be specified)

(Add, search, change, delete, etc.) *The format of some SQL statements differs depending on the RDB server.



データベースとKVの接続の仕組み

KV-XCM02(PLC):RDB client and database: RDB server sends and receives **SQL statements** via **LAN cable (TCP/IP)** . This document explains how to (1) set up the RDB server (database) and (2) configure communication settings in order to connect the KV-XCM02 to the database.



データベースの管理ツールについて

The RDB server itself is an application software that has a mechanism for storing and managing data.

It does not have a user interface.

Users now use tools such as Notepad and Excel to view and edit CSV files.

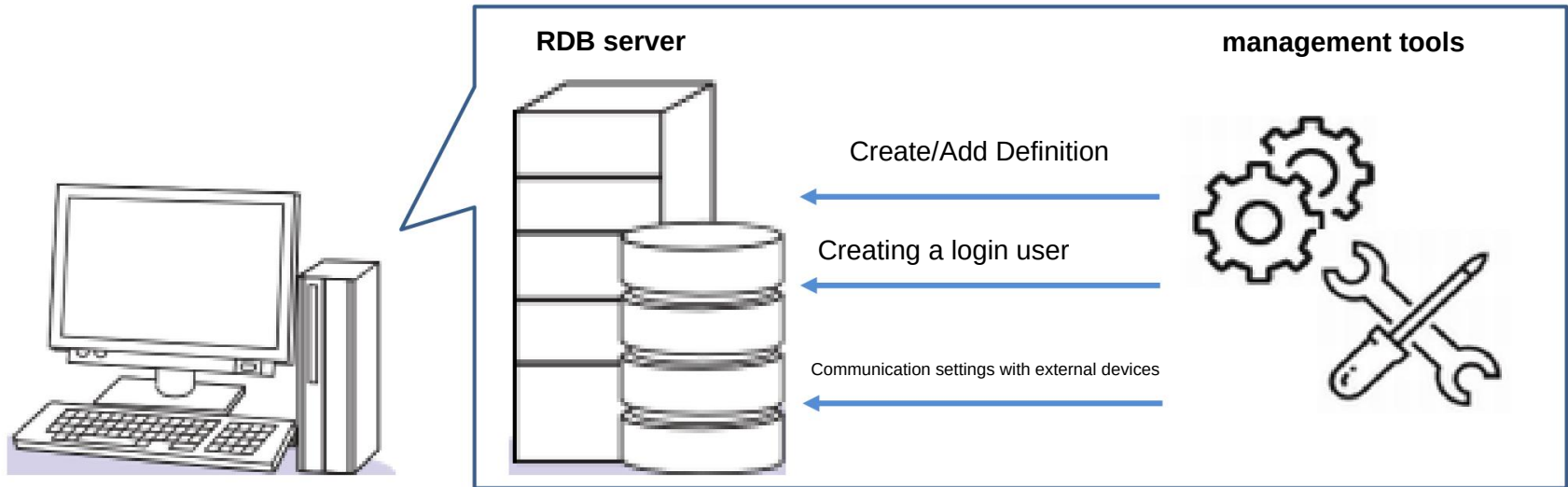
Management tools are also available for viewing and editing the database.

In some cases, it is installed at the same time as the RDB server installation, but

In some cases, separate installation may be required.

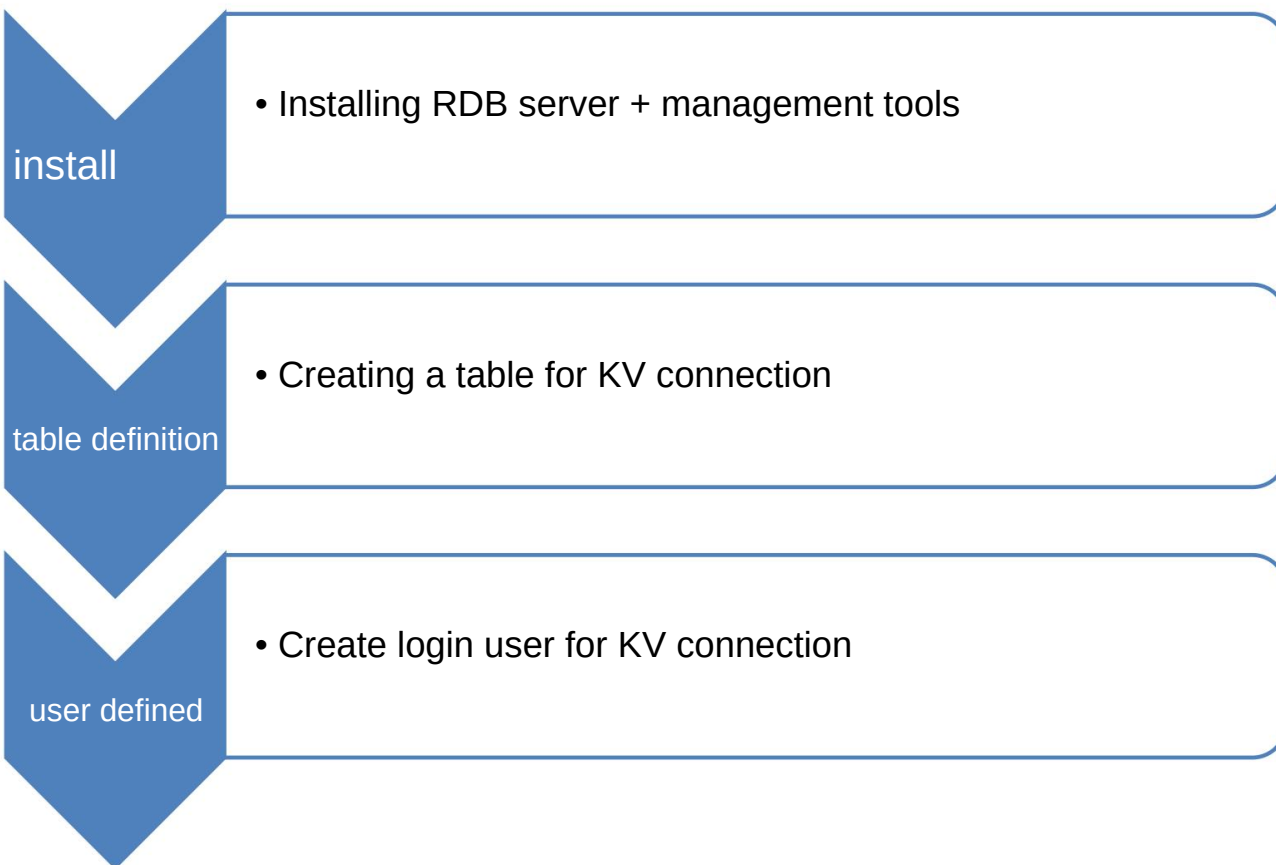
In this manual, settings are performed using management tools.

RDB server	management tools
PostgreSQL	pgAdmin
Microsoft SQL Server	SQL Server Management Studio
MySQL	MySQL Workbench
Oracle Database	SQL Developer



RDBサーバのセットアップ[®]

Follow the steps below to set up.



RDBサーバのセットアップ ~RDBサーバと管理ツールのインストール~

- RDB server: MySQL server
- Management tool: MySQL Workbench

ダウンロード

install

Create table

Creating a user

We will explain how to download MySQL and install the RDB server and management tools. [Access <https://dev.mysql.com/downloads/windows/installer/8.0.html> and select \[Download\] below.](https://dev.mysql.com/downloads/windows/installer/8.0.html) The page will change.

MySQL Community Downloads

MySQL Installer

General Availability (GA) ReleasesArchives

MySQL Installer 8.0.20

Select Operating System:
Microsoft Windows

Looking for previous versions?choice

Windows (x86, 32-bit), MSI Installer (mysql-installer-web-community-8.0.20.0.msi)	8.0.20	24.4M	Download MD5: 26ae47807122bf0052b99ebf893b0dac Signature
Windows (x86, 32-bit), MSI Installer (mysql-installer-community-8.0.20.0.msi)	8.0.20	420.6M	Download MD5: a69c77fe787654d8931079b4823b8e1a Signature

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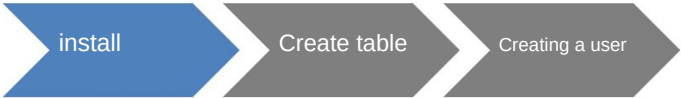
We suggest that you use the MD5 checksums and GnuPG signatures to verify the integrity of the packages you download.

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ダウンロード



The login screen for your MySQL.com account will be displayed, but this time we will download without registering an account, so select "No thanks, just start my downloads". The installer will begin downloading. After the download is complete, launch the installer.

MySQL Community Downloads

Login Now or Sign Up for a free account.

An Oracle Web Account provides you with the following advantages:

- Fast access to MySQL software downloads
- Download technical White Papers and Presentations
- Post messages in the MySQL Discussion Forums
- Report and track bugs in the MySQL bug system

Login »
using my Oracle Web account

Sign Up »
for an Oracle Web account

MySQL.com is using Oracle SSO for authentication. If you already have an Oracle Web account, click the Login link. Otherwise, you can signup for a free account by clicking the Sign Up link and following the instructions.

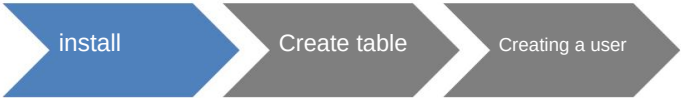
No thanks, just start my download.

choice

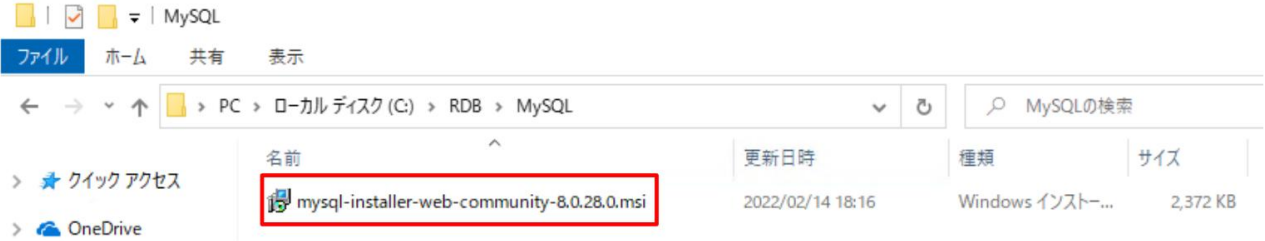
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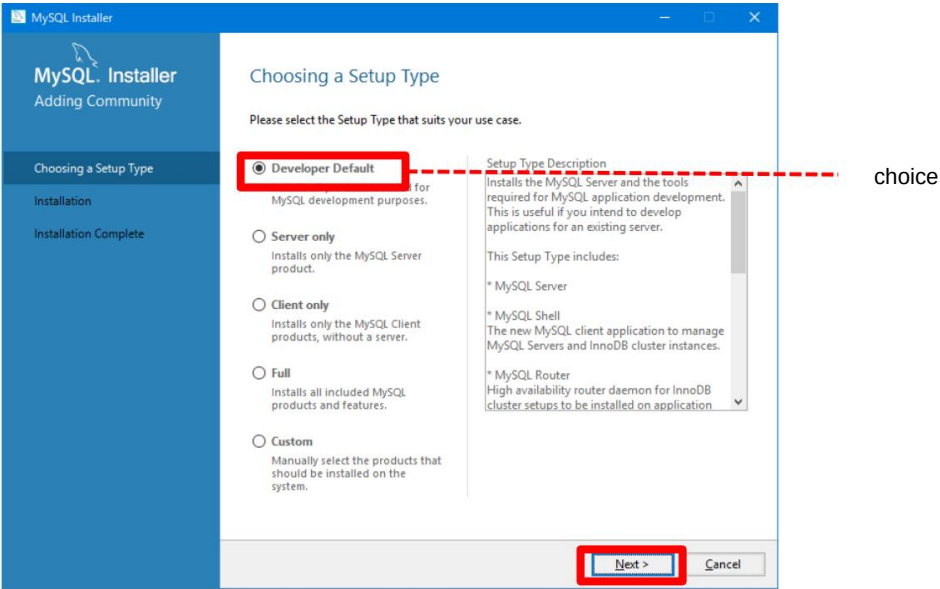
インストール



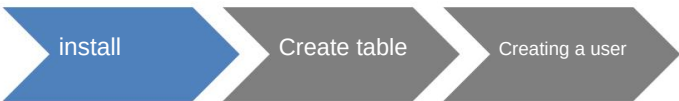
Double-click the downloaded installer (msi file) to start it.



When the installer starts, select the setup method. Select “Developer Default” which allows you to install the basic set, and then select [Next].



インストール

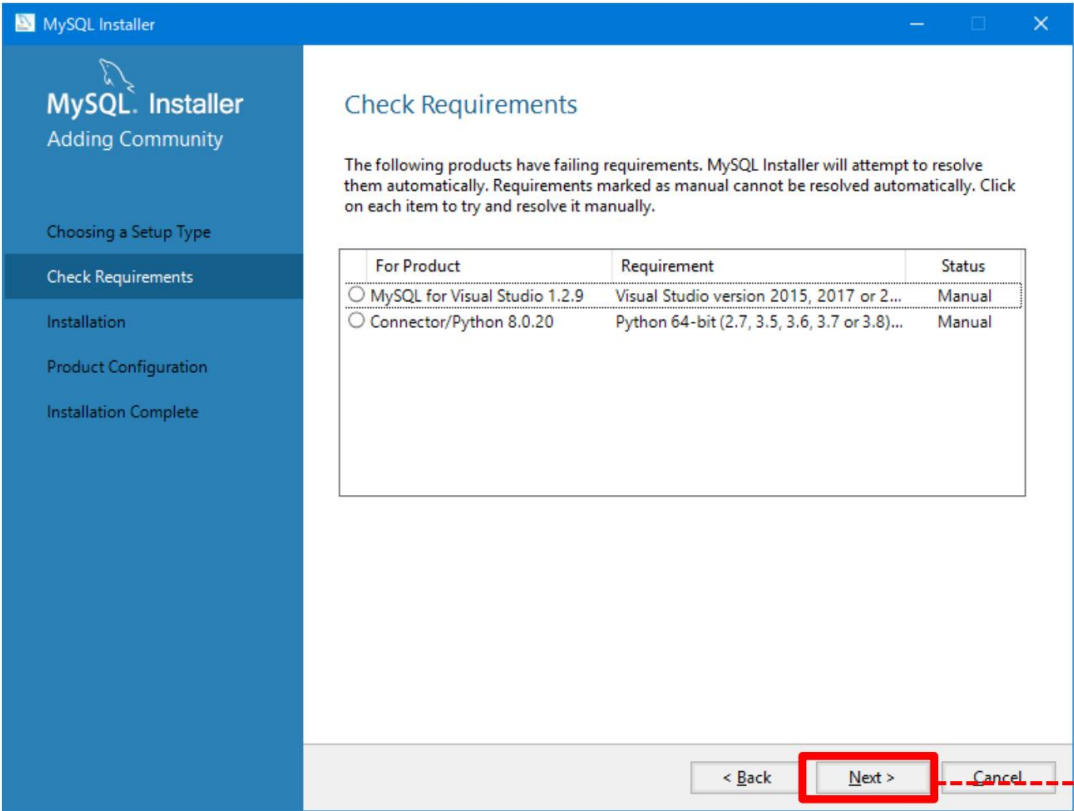


A warning will be displayed stating that "Some of MySQL cannot be installed because the requirements are insufficient to install it."

If the Status is "blank", MySQL will automatically resolve it during installation, but Items whose Status is displayed as "Manual" cannot be resolved automatically.

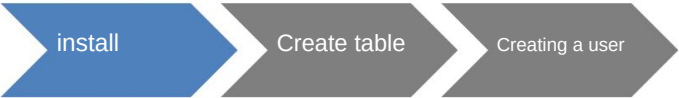
Please check the RDB server manual and take action as necessary.

You will receive a warning from the program development software such as Visual Studio or Python that the requirements for connecting to MySQL are not met, but this is not necessary for connecting to KV, so select [Next] here.

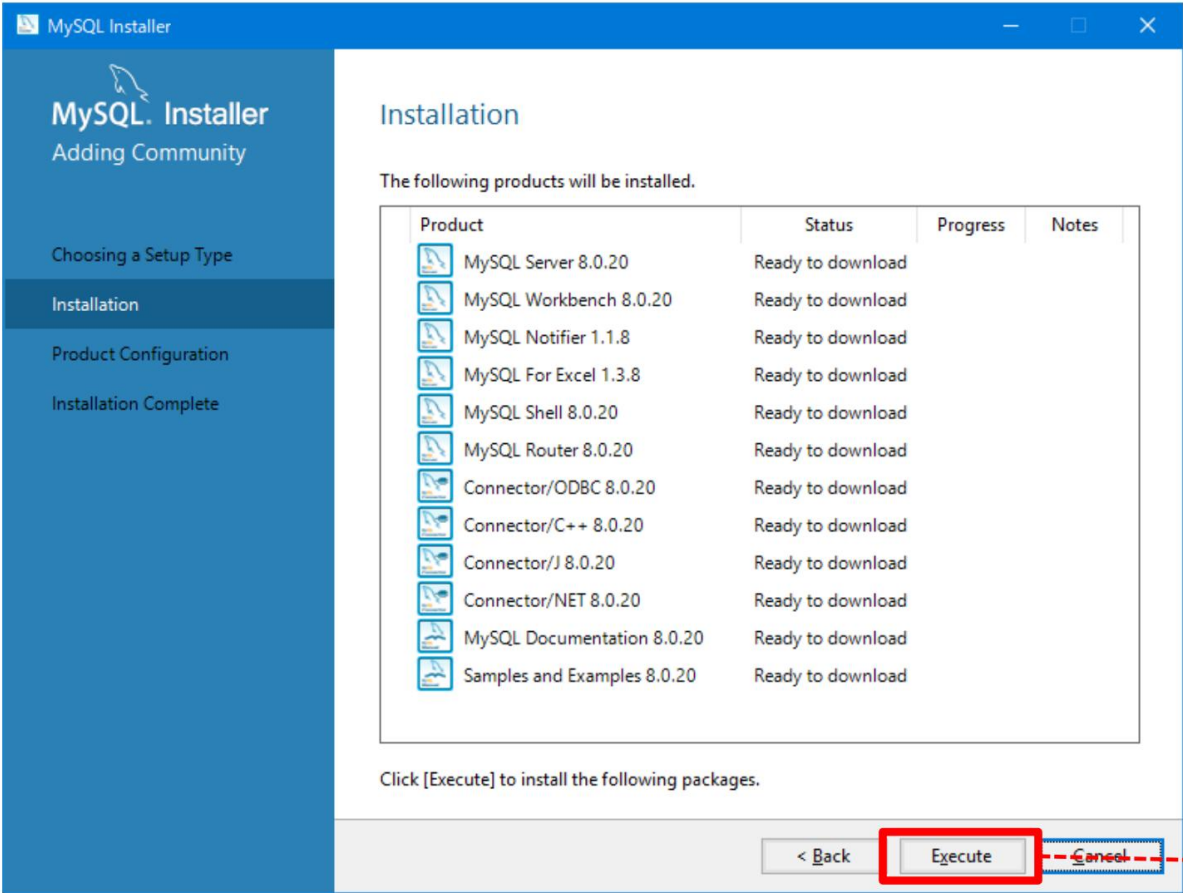


choice

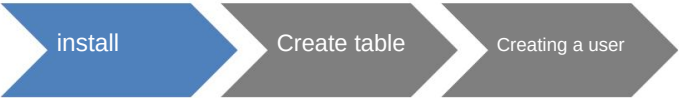
インストール



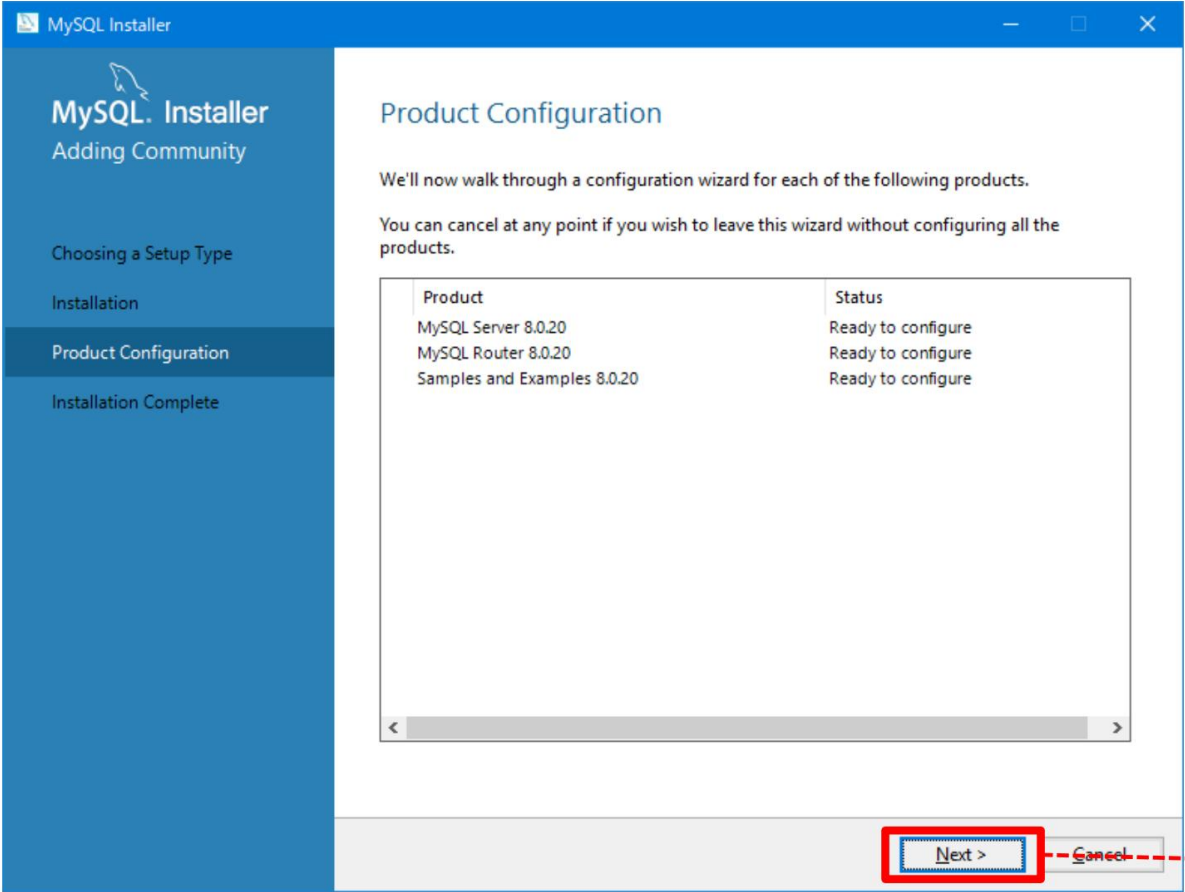
A list of installation tools will be displayed.
You can confirm that the RDB server "MySQL" and the management tool "Workbench" are installed.
Select [Execute] to begin the installation.



インストール



Select Next.



インストール

install

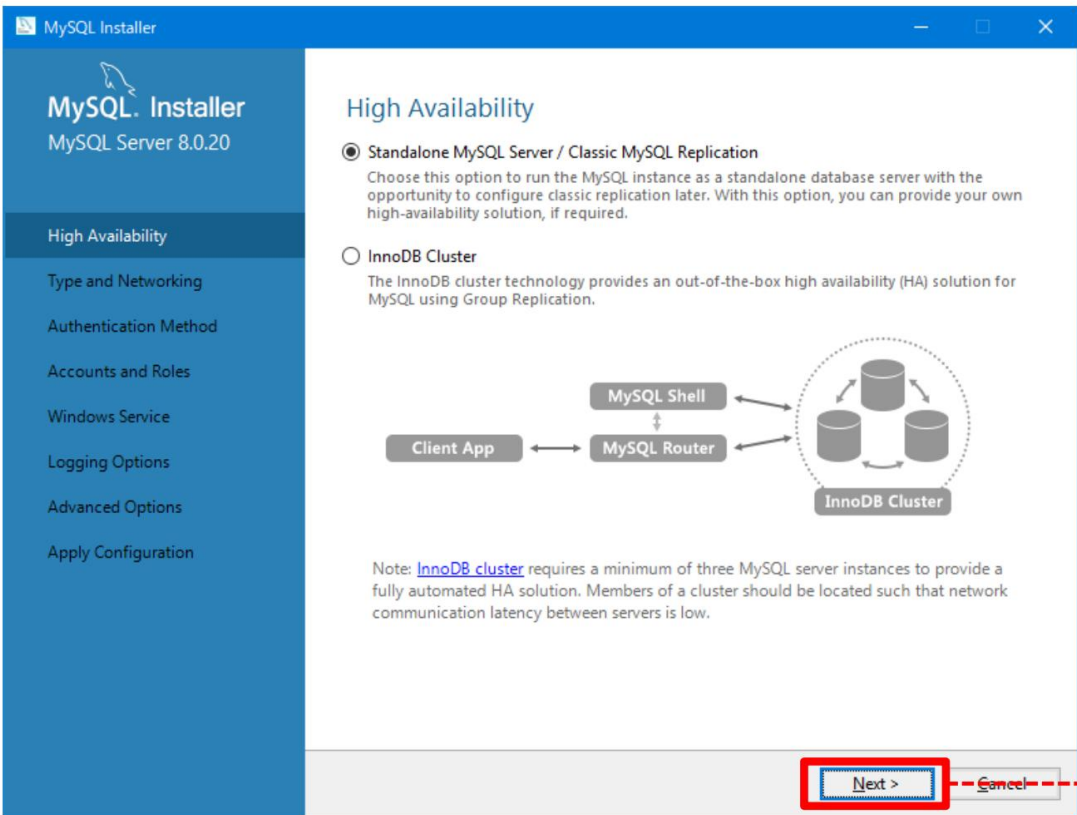
Create table

Creating a user

This is an extension that operates using multiple MySQL servers. You can also set it later. Since no extensions are required this time, select "Standalone MySQL Server / Classic MySQL Replication" to operate independently. For details on

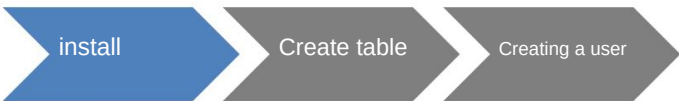
this setting, please check the RDB server manual.

Select Next.



choice

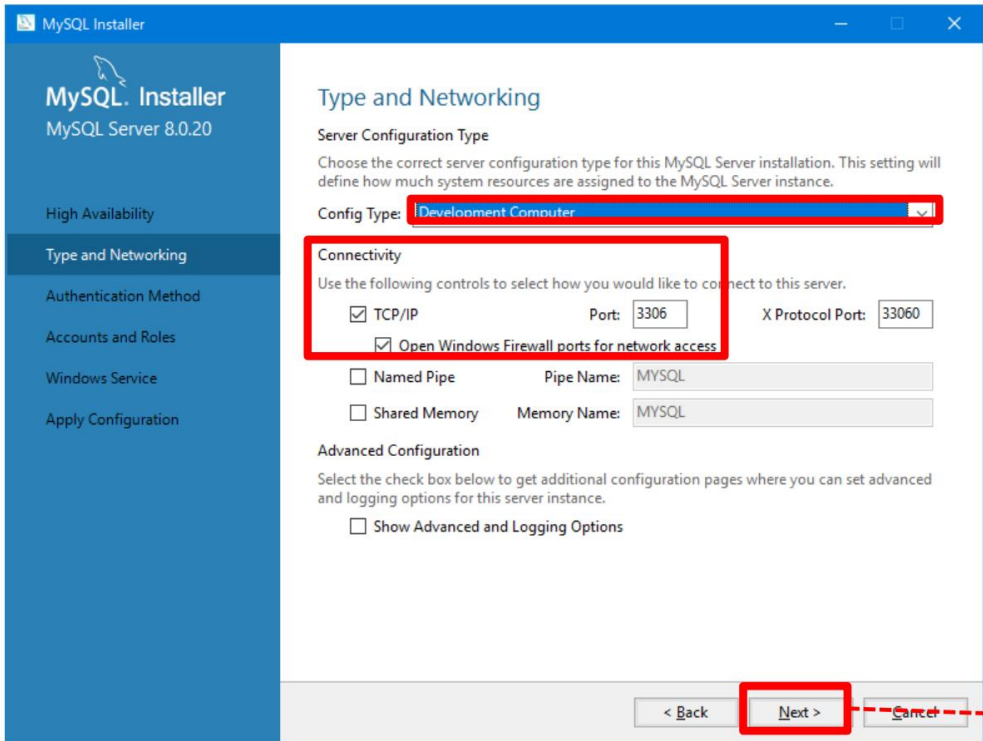
インストール



Configure the usage settings and network settings for the RDB server. This time we will be testing the connection with the KV, so select "Development Computer". For other usage settings, please check the RDB server manual and select according to the situation.

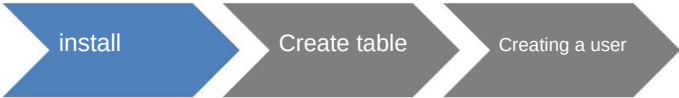
Regarding the network settings, put a $\checkmark$ in the "TCP/IP" box required for connection with KV and set the port number. Default is "3306". You can also configure the Windows firewall, so

Enter $\checkmark$ in "Open Windows Firewall ports network access".
Select Next.



choice

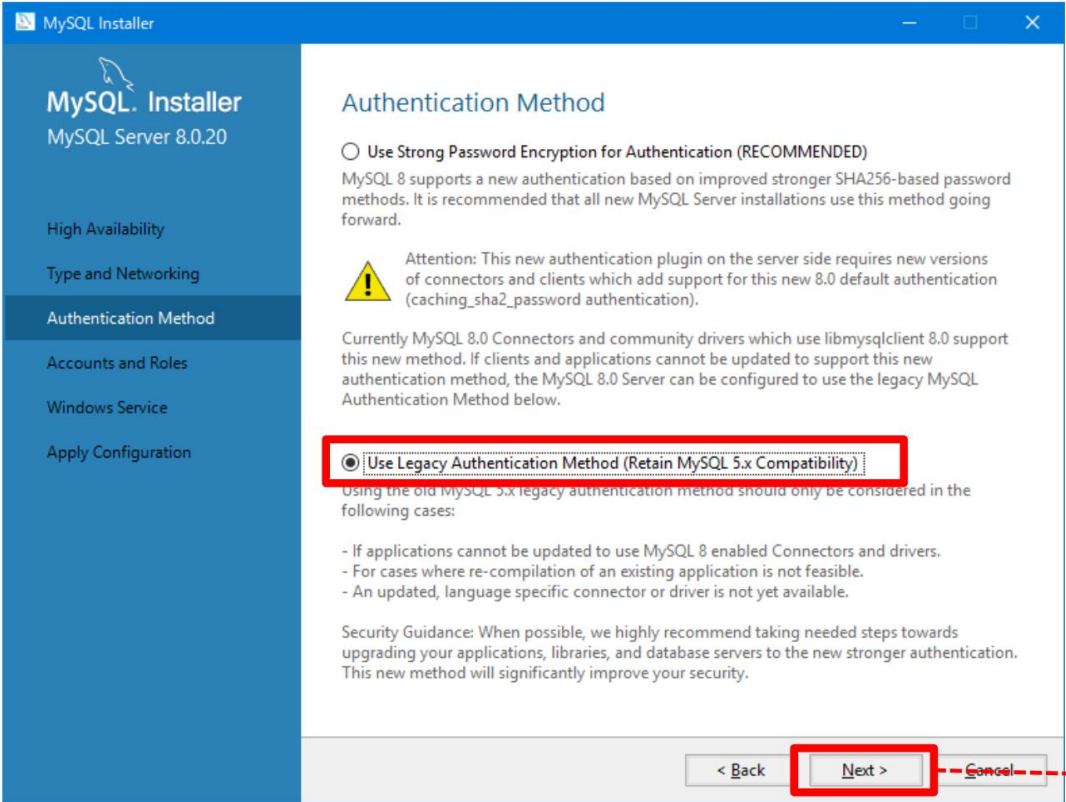
インストール



Select the password authentication method.

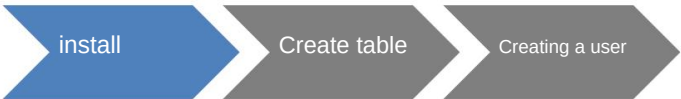
KV supports the "Standard" authentication method. Select **Use Legacy Authentication Method**. *If you select and install the "SHA-256" authentication method, you will need to make changes in the "Communication Settings" configuration file in this document.

Select Next.



choice

インストール



Set the password for the administrative user "root", which exists by default.

*Please keep this in mind as it will be required to log in to the RDB server. You can also create a login user to use. (You can also create one later.) Select [Next].

MySQL Installer
MySQL Server 8.0.20

High Availability
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Apply Configuration

Accounts and Roles

Root Account Password
Enter the password for the root account. Please remember to store this password in a secure place.

MySQL Root Password:

Repeat Password:

Password strength: **Weak**

MySQL User Accounts
Create MySQL user accounts for your users and applications. Assign a role to the user that consists of a set of privileges.

MySQL User Name	Host	User Role
 KV	%	User Admin

Add User (optional)
Edit User
Delete

< Back **Next >** Cancel

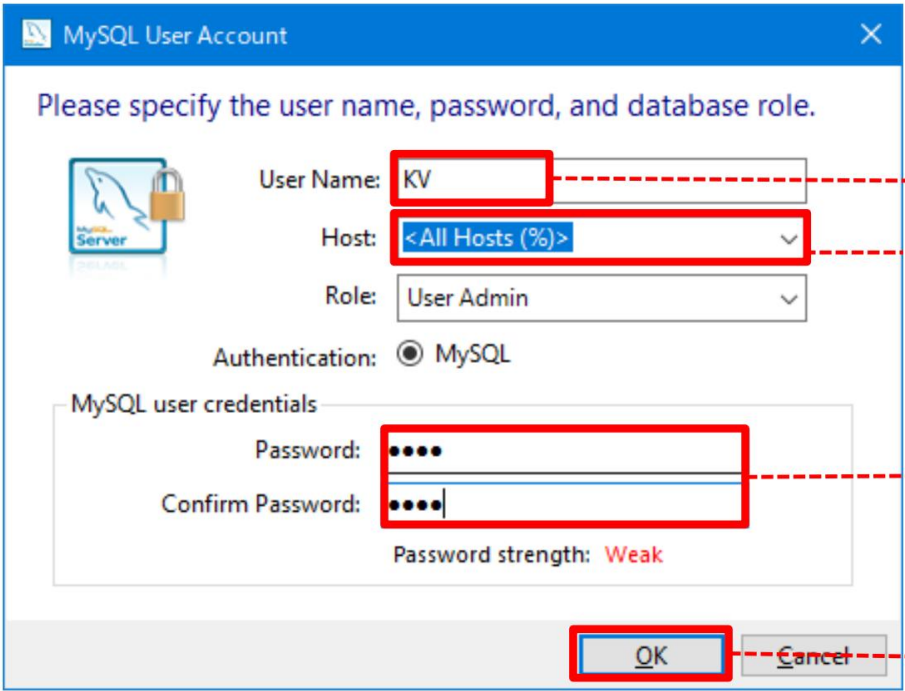
インストール

install

Create table

Creating a user

``Option``: When creating a user, you can create a login user in the window below.



The image shows a 'MySQL User Account' dialog box with the following fields and options:

- User Name:** A text input field containing 'KV'.
- Host:** A dropdown menu showing '<All Hosts (%)>'.
- Role:** A dropdown menu showing 'User Admin'.
- Authentication:** A radio button selected for 'MySQL'.
- MySQL user credentials:** A section containing:
 - Password:** A text input field with masked characters (dots).
 - Confirm Password:** A text input field with masked characters (dots).
 - Password strength:** A label indicating 'Weak'.
- Buttons:** 'OK' and 'Cancel' buttons at the bottom right.

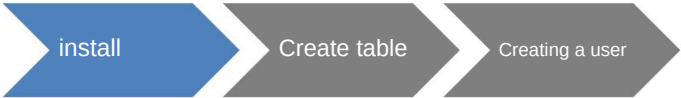
Username input

Specify connectable IP addresses
"%" means all IP addresses.

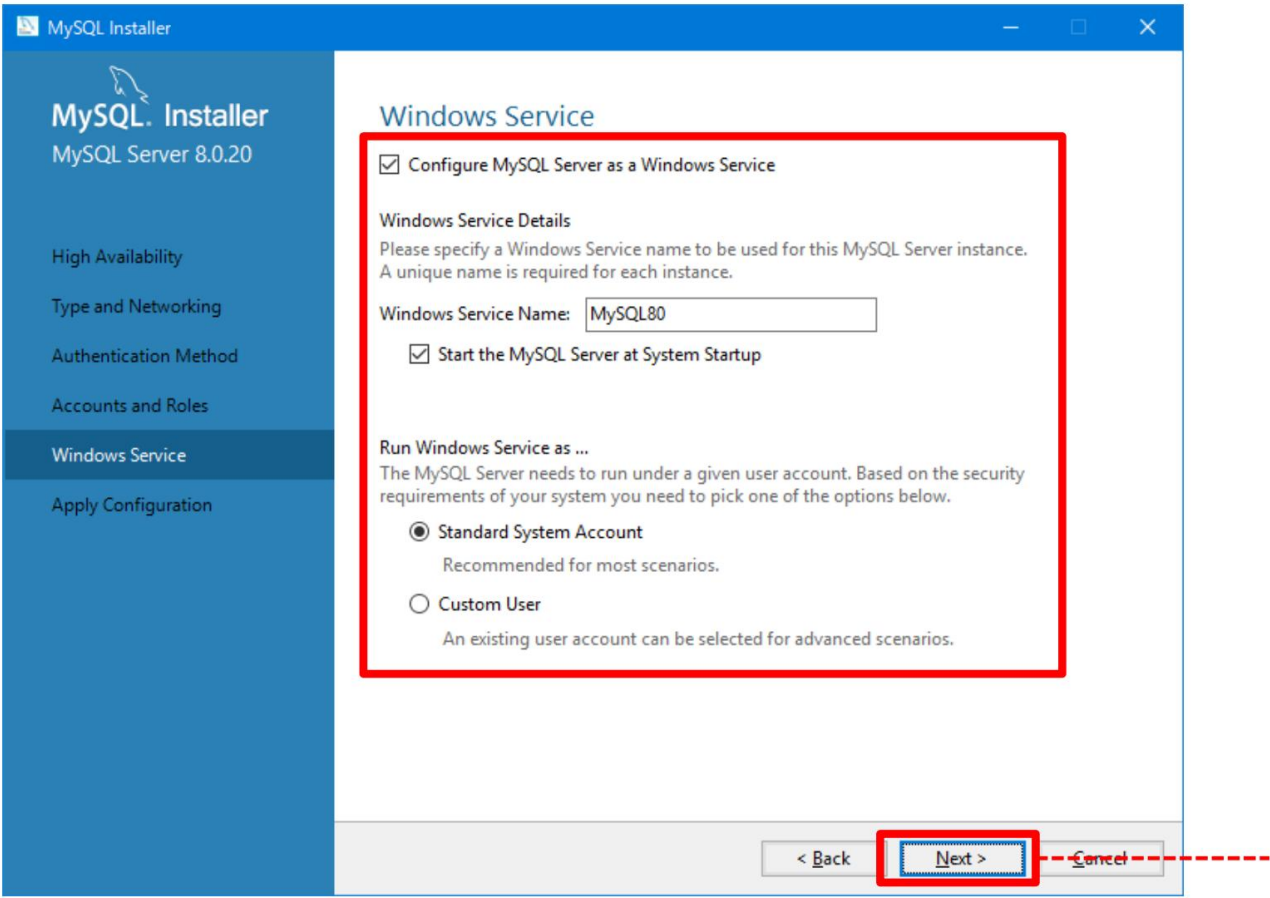
enter password

choice

インストール

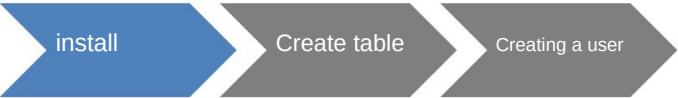


You can register MySQL as a Windows feature. You can also start the MySQL server when you start your PC.
If you select "**Custom** User", you can set up a user on Windows who can operate the RDB server service. The following are recommended settings. After setting, select [Next].

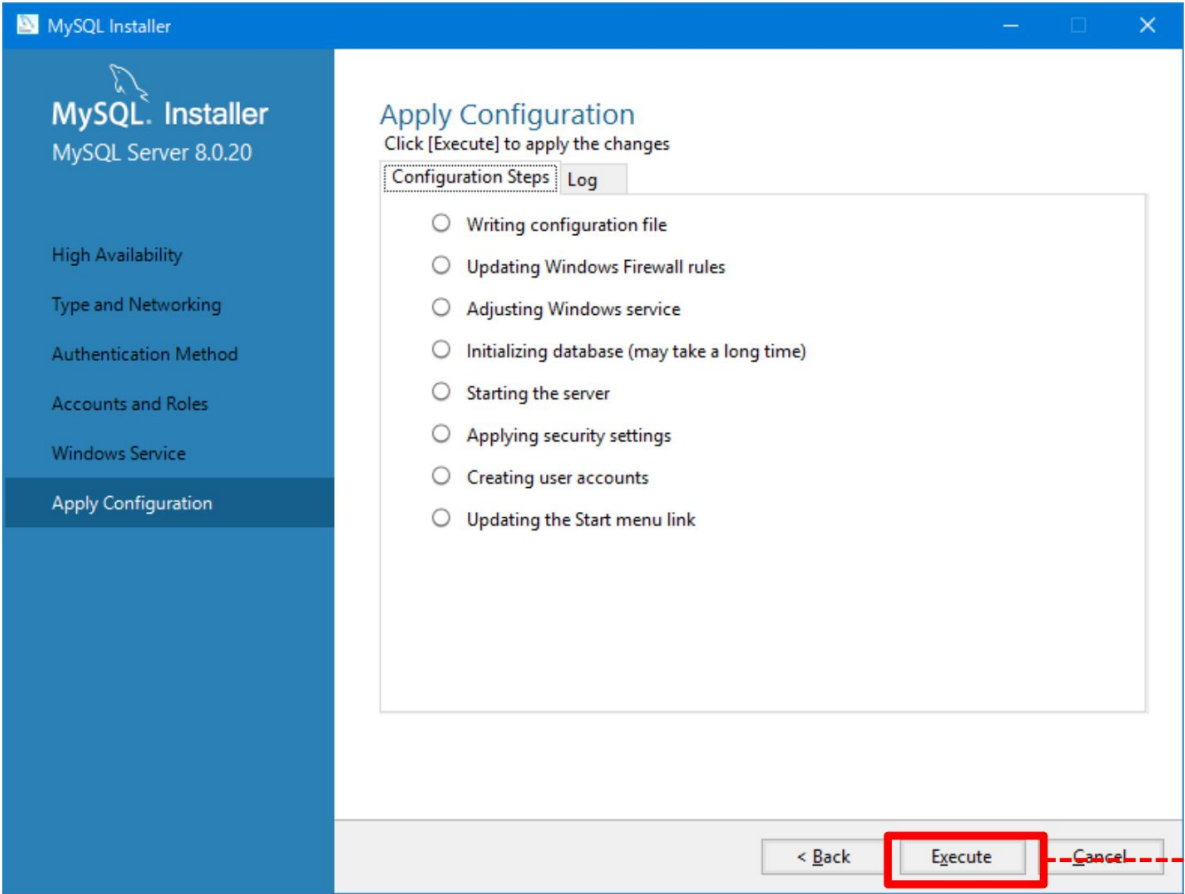


choice

インストール

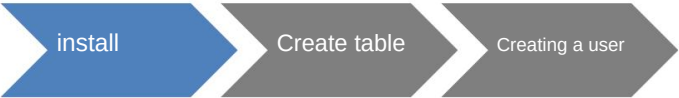


Select [Execute] to apply the settings.

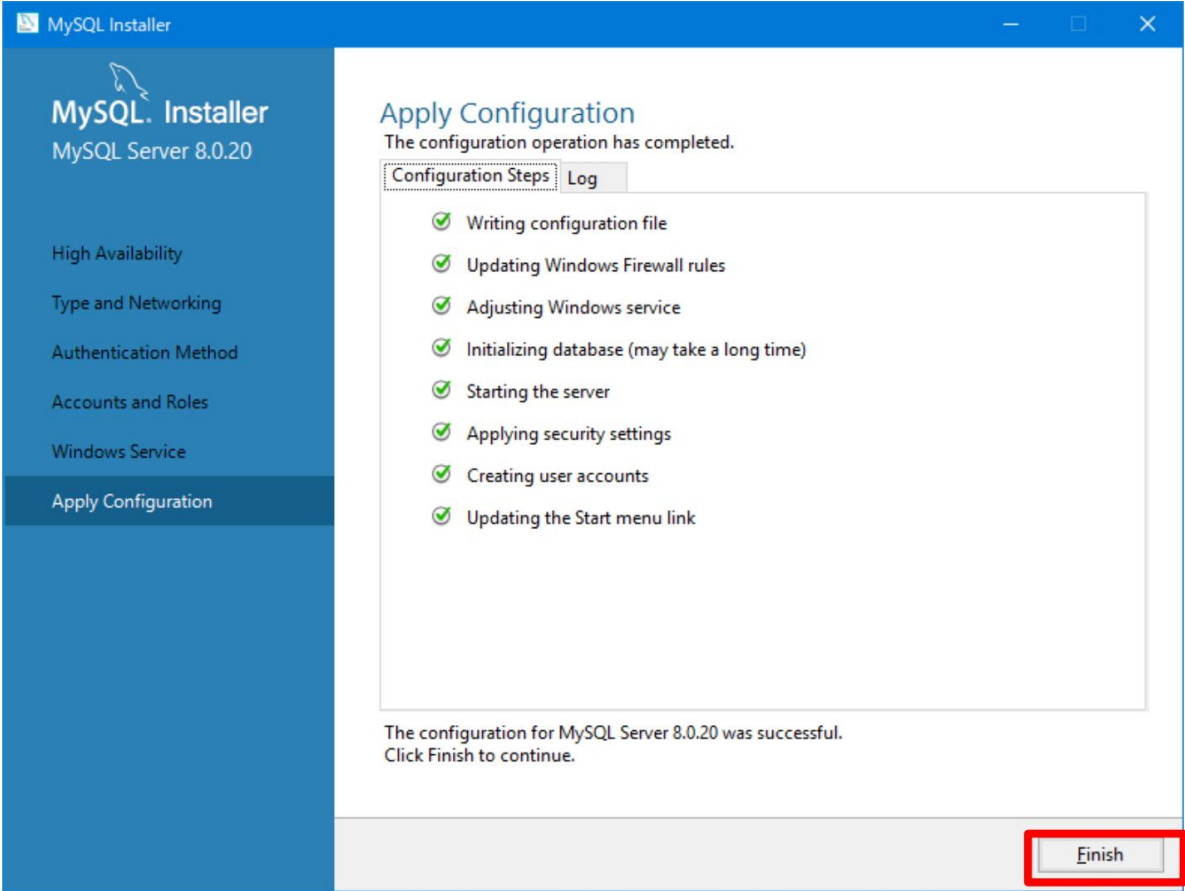


choice

インストール

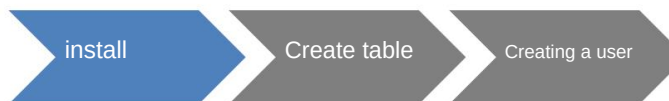


After reflecting, select [Finish].



choice

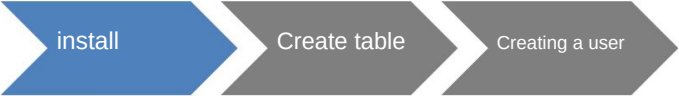
インストール



This is an extended function that performs load balancing by using multiple RDB servers. This is also not necessary for this connection test, so select [Finish] without making any settings. Please check the RDB server manual for the setting method.

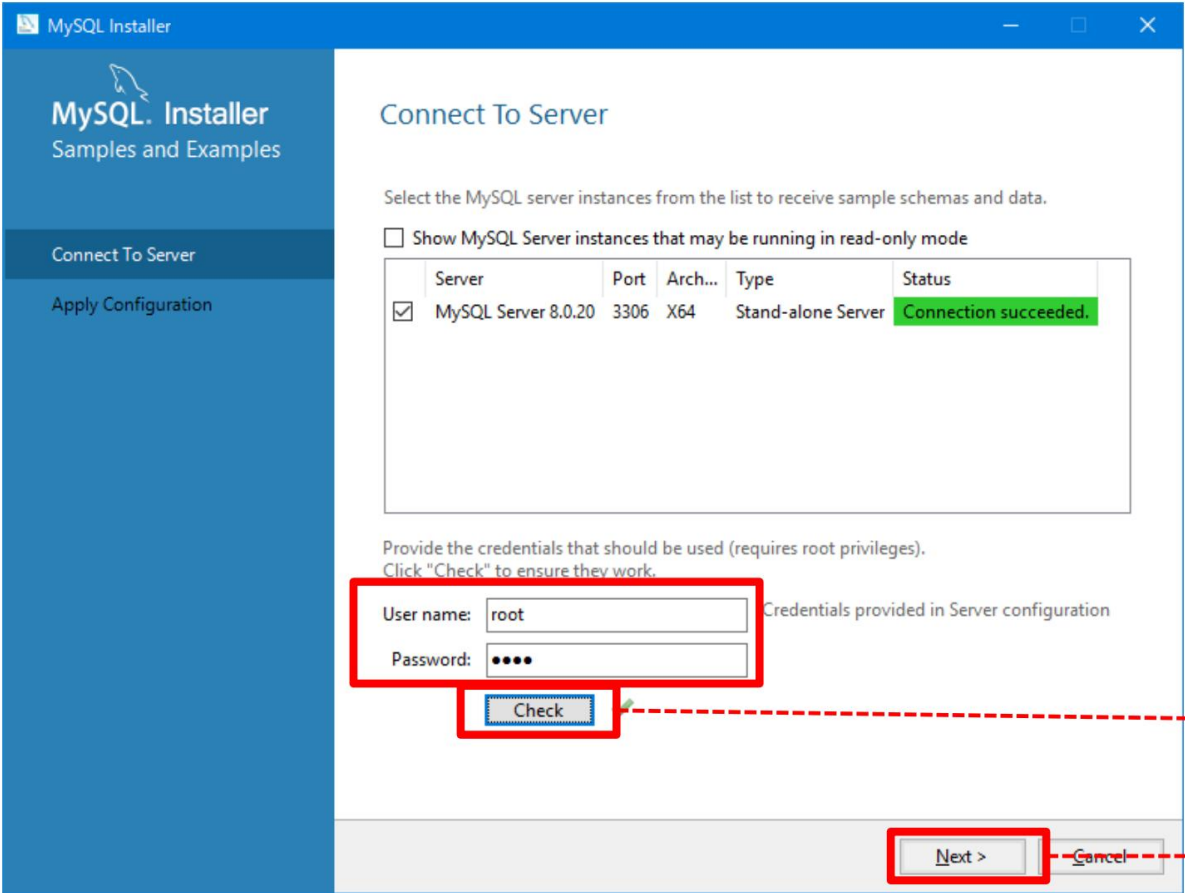
choice

インストール



Check the connection.

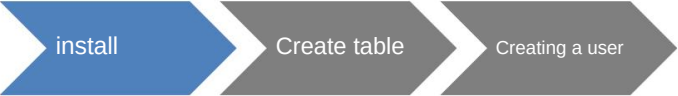
Enter the account information set on page 21 (administrative user name is "root") and select [Check] to verify. Once the connection is successful, you will be able to select Next.



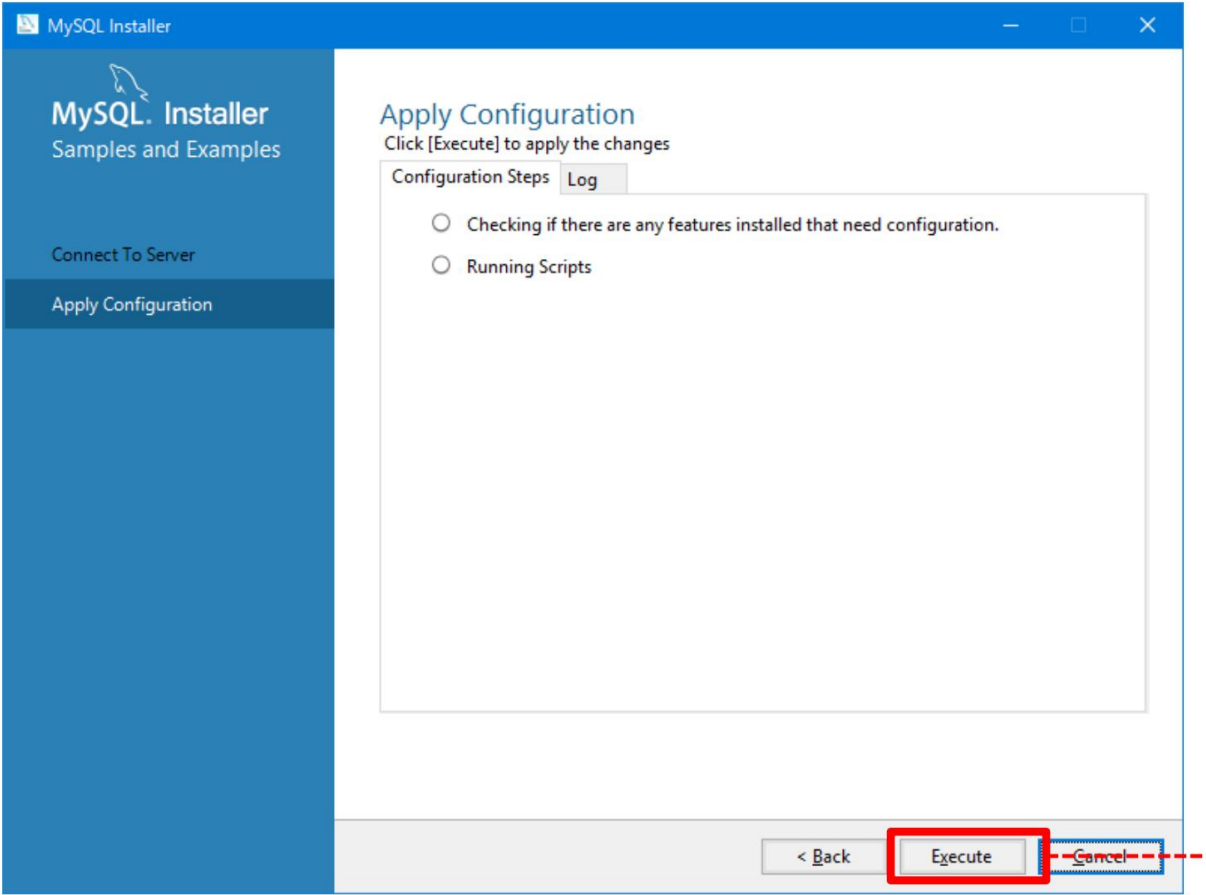
choice

choice

インストール

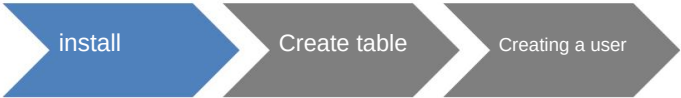


Reflect the settings. Select Execute.

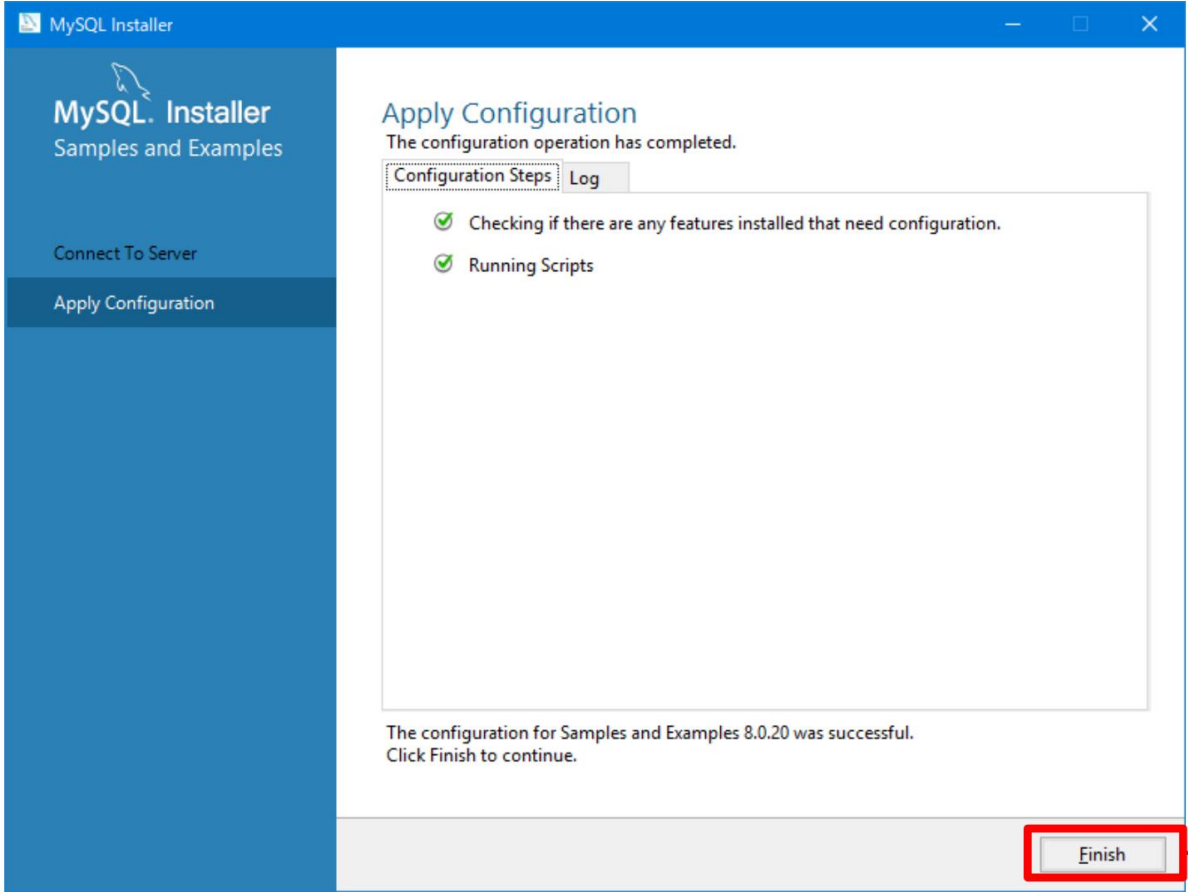


choice

インストール

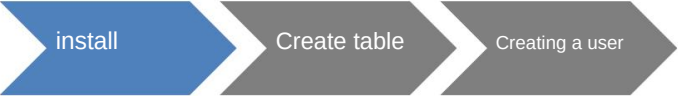


After applying the settings, select [Finish].

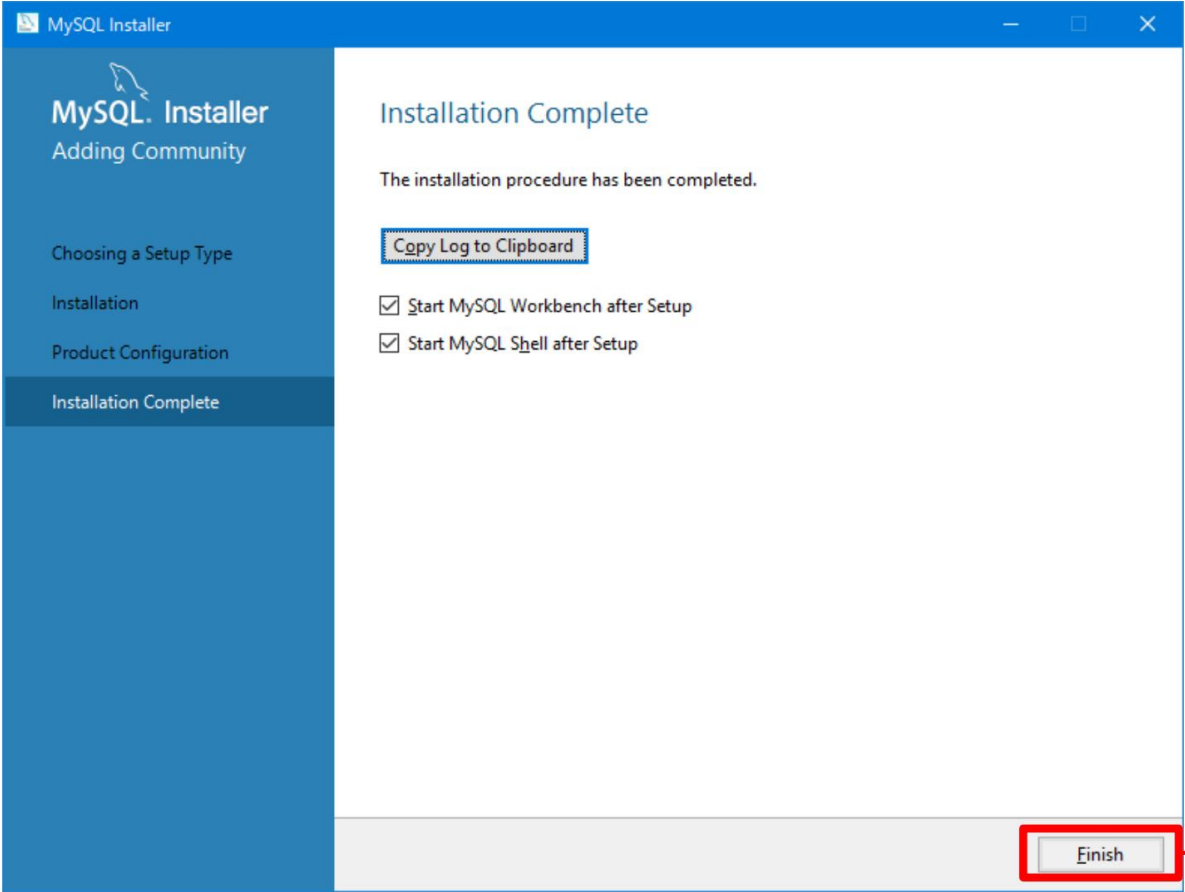


choice

インストール



You can output the installation settings if necessary.
Select Finish.



choice

RDBサーバのセットアップ ～KV接続用テーブルの作成～

- Set up the RDB server using management tools

install

Create table

Creating a user

テーブル作成について

Create a table on the RDB server. MySQL has the following hierarchy.

Schema > Tables (column data types)

You can define any name for each. Set the column data type according to your purpose.

For details, please check the RDB server manual.

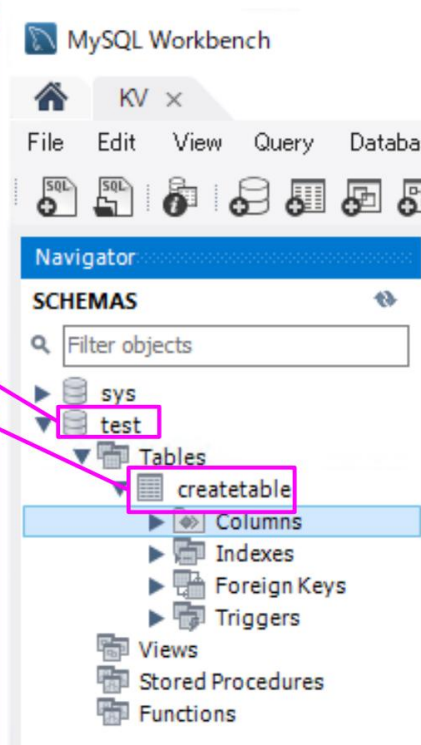
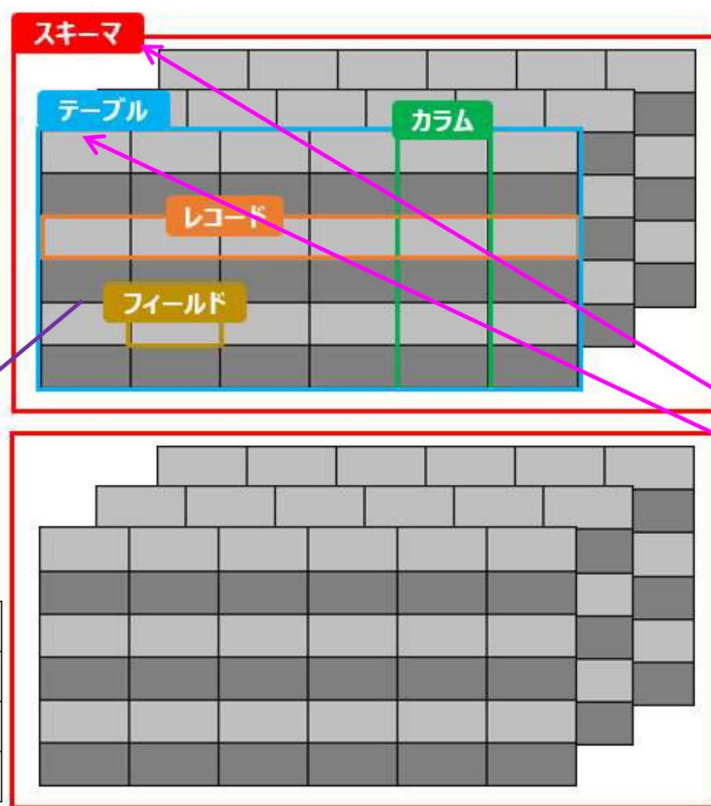


Table design example

column	data type
serial string type	
Measurement value	Floating point real number
Registration date	date type

install

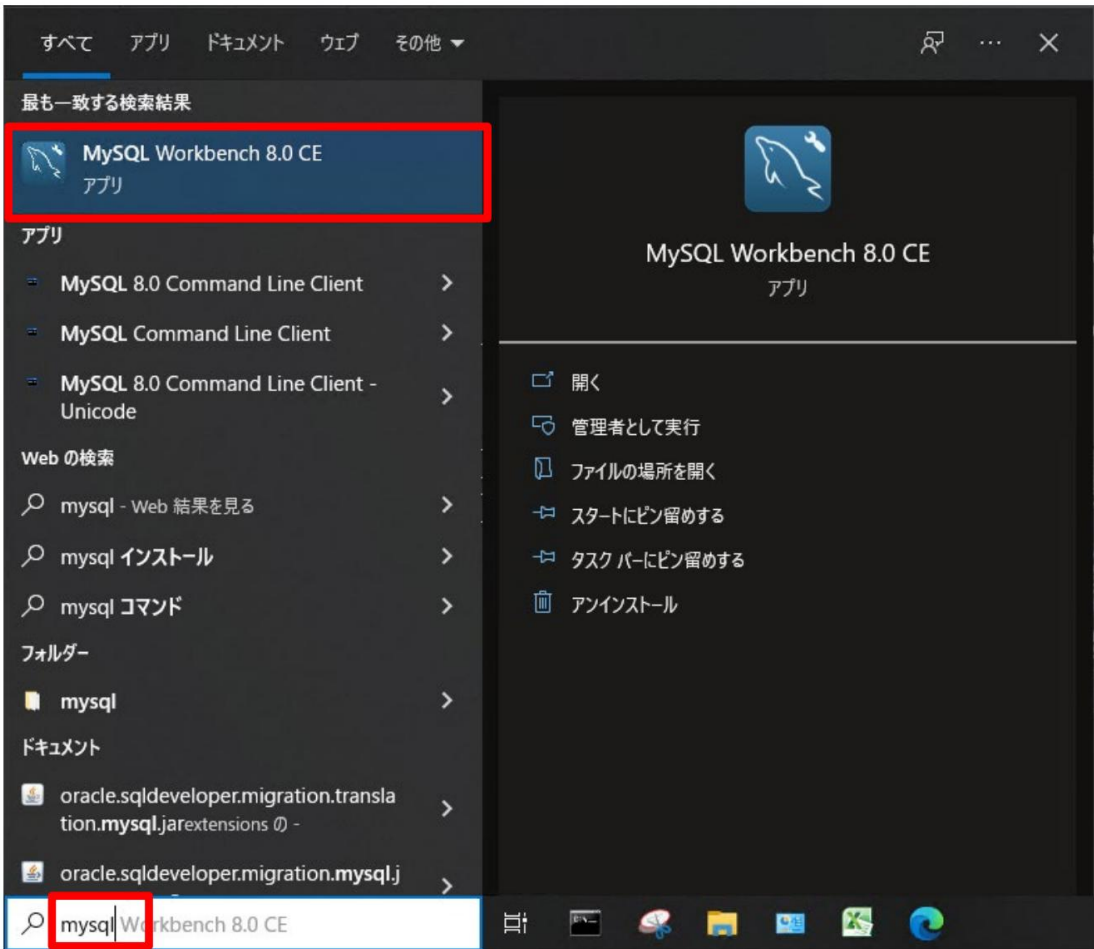
Create table

Creating a user

MySQL Workbenchの起動

Start the installed MySQL Workbench (MySQL management tool).

You can search for the target and start it by typing something like "mysql".



接続の作成

install

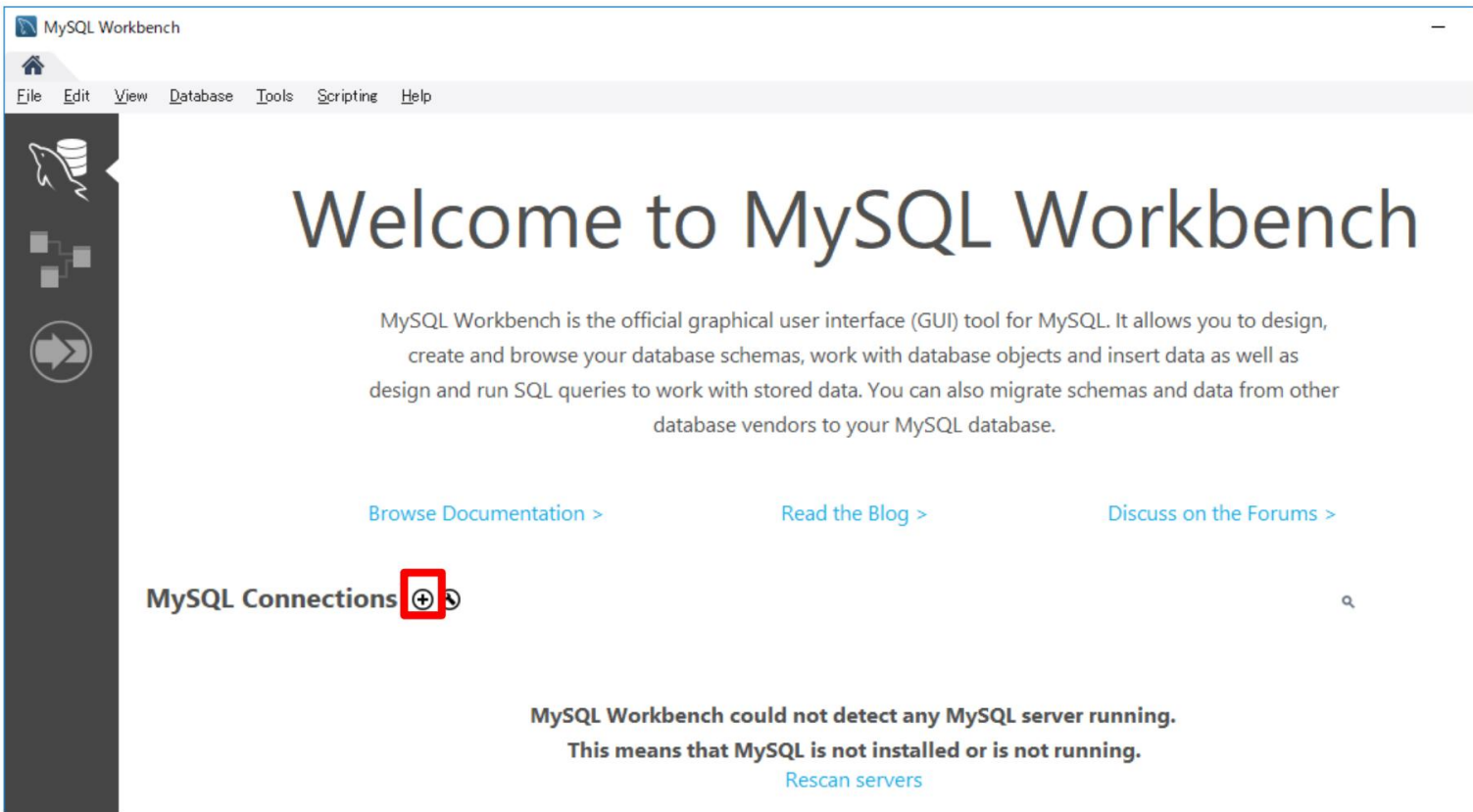
Create table

Creating a user

Create connection settings to the RDB server from MySQL Workbench.

Once you create a connection setting, you can connect to the RDB server from the next time using the connection setting you created.

Click the "+" button on the right side of "MySQL Connections".



接続の作成

install

Create table

Creating a user

A connection settings window will appear. Create a connection as an administrative user.

Connection Name is optional, enter "127.0.0.1" (*) for Hostname, and "root" for Username. Enter the port and password set during installation. (Refer to the setting procedure on page 20) *This is a special IP address that points to your own device. You can also enter the IP address of your own device.

Setup New Connection

Connection Name: Type a name for the connection

Connection Method: Method to use to connect to the RDBMS

Parameters

Hostname: Port: Name or IP address of the server host - and TCP/IP port.

Username: Name of the user to connect with.

Password: The user's password. Will be requested later if it's not set.

Default Schema: The schema to use as default schema. Leave blank to select it later.

接続の作成

install

Create table

Creating a user

Select the SSL tab.

SSL : Abbreviation for Secure Sockets Layer, a mechanism for encrypting communications. KV is not supported. You can test the connection by changing Use SSL to "No" and selecting "Test Connection".

The screenshot shows the 'Setup New Connection' dialog box. The 'Connection Name' is 'test_connect'. The 'Connection Method' is 'Standard (TCP/IP)'. The 'SSL' tab is selected. The 'Use SSL' dropdown is set to 'No'. The 'Test Connection' button is highlighted with a red box.

Setup New Connection

Connection Name: test_connect Type a name for the connection

Connection Method: Standard (TCP/IP) Method to use to connect to the RDBMS

Parameters **SSL** Advanced

Use SSL No Turns on SSL encryption. Connection will fail if SSL is not available.

SSL Key File: Path to Client Key file for SSL.

SSL CERT File: Path to Client Certificate file for SSL.

SSL CA File: Path to Certificate Authority file for SSL.

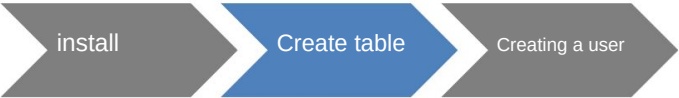
SSL Cipher: Optional : separated list of permissible ciphers to use for SSL encryption.

SSL Wizard ...

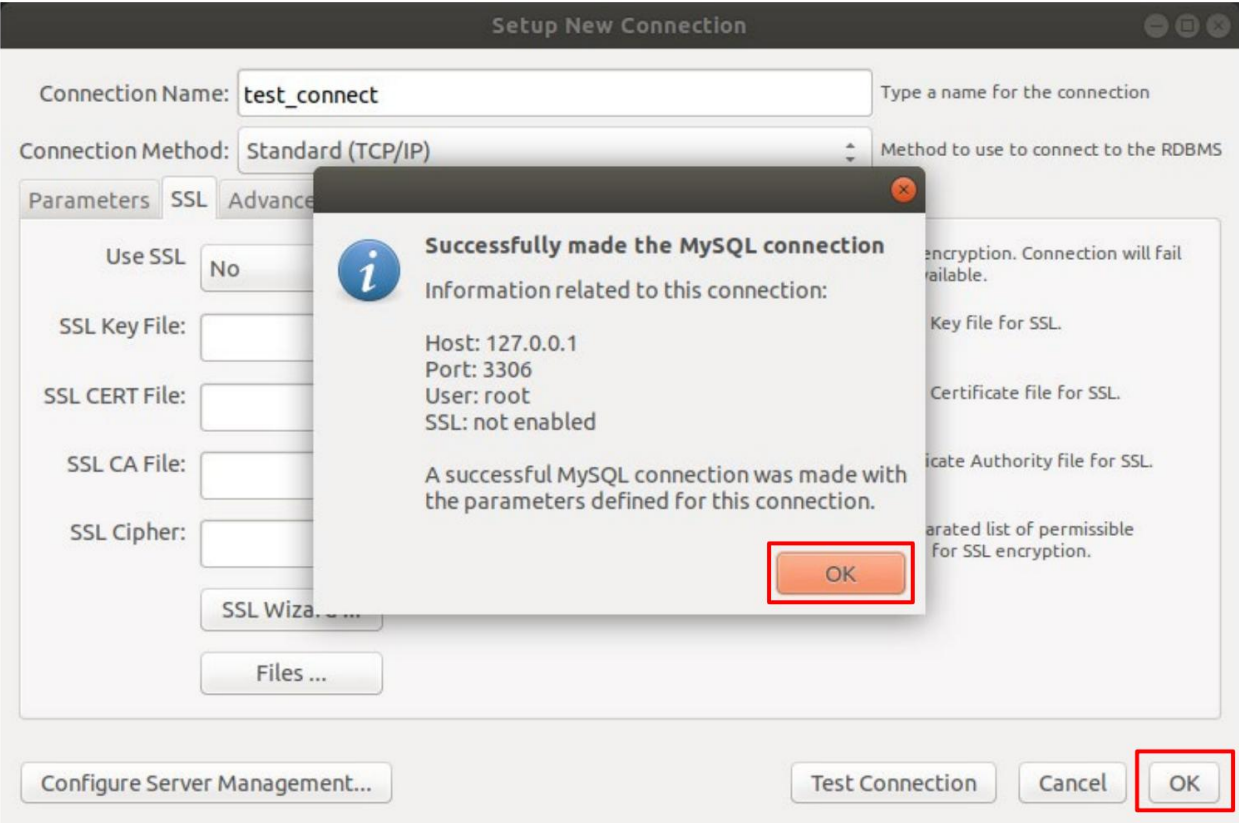
Files ...

Configure Server Management... Test Connection Cancel OK

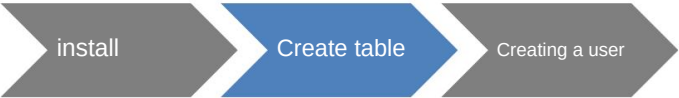
接続の作成



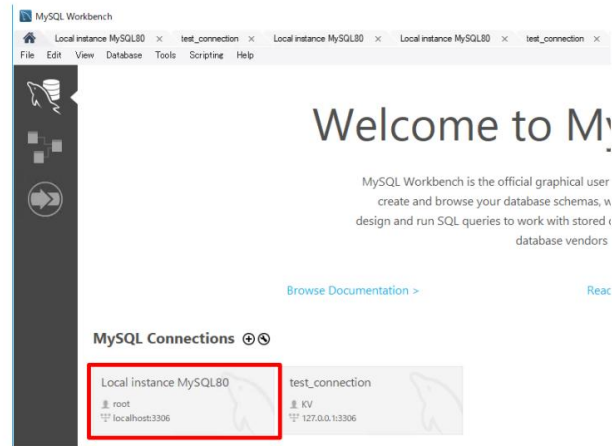
After entering the password, a connection test result dialog will be displayed.
If you are satisfied, select OK to create the connection.



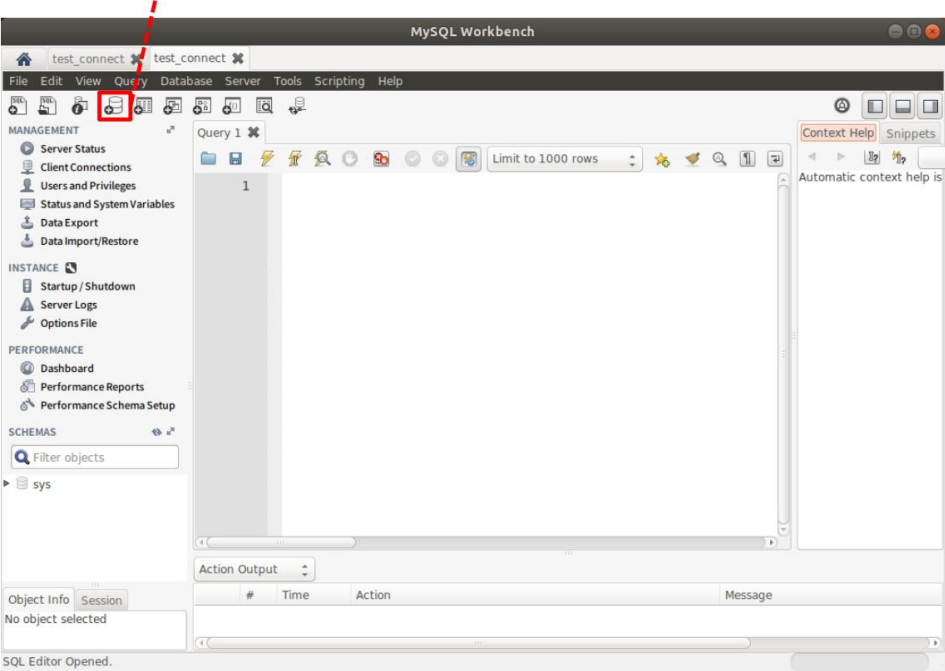
スキーマの作成



Log in to the RDB server using the connection settings you created.



Create a schema. Select
Create a new schema in the
connected server at the bottom of the menu bar.



スキーマの作成

install

Create table

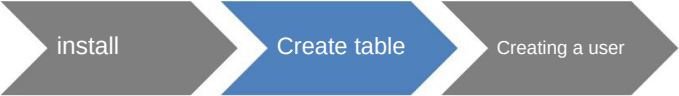
Creating a user

The schema creation screen will be displayed.

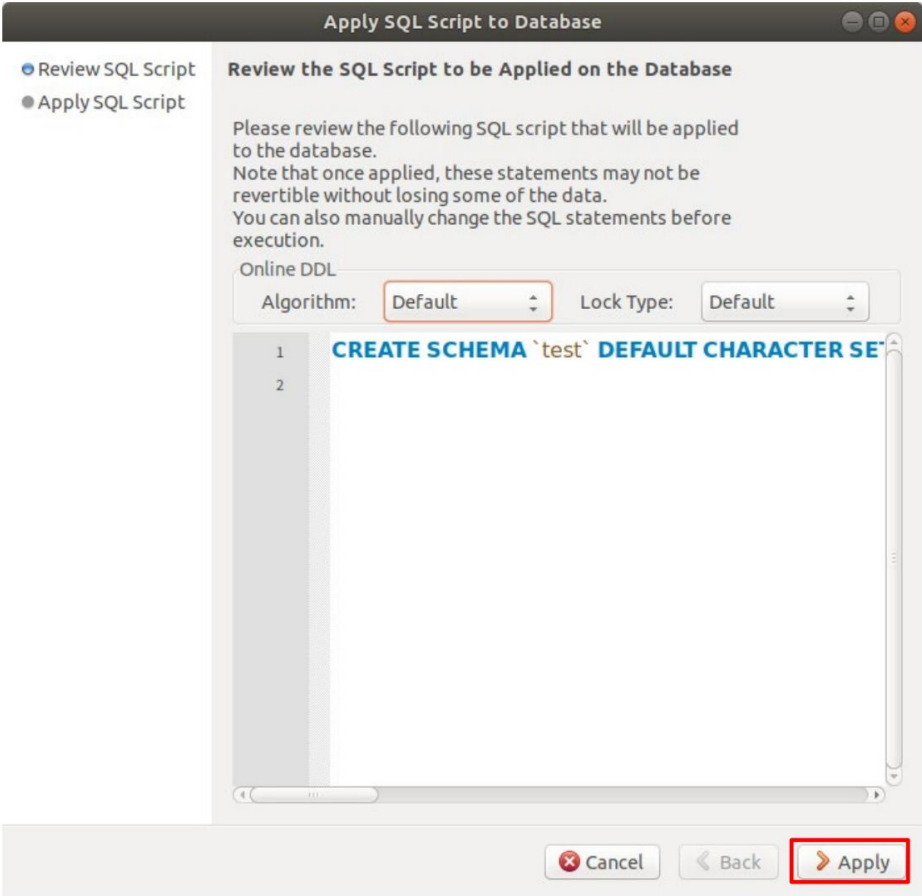
Select the schema name (in this case test) for Name, “utf8 – utf8_bin” for Default Collation (character code),
Select Apply.

The screenshot shows a dialog box titled "new_schema - Schema". It has a "Name" field containing "test" and a "Default Collation" dropdown menu set to "utf8 - utf8_bin". A "Rename References" button is located below the "Name" field. To the right of the "Name" field, there is explanatory text: "The name of the schema. It is recommended to use only alpha-numeric characters. Spaces should be avoided and be replaced by _". Below the "Default Collation" dropdown, there is more text: "Specifies which charset/collation the schema's tables will use if they do not have an explicit setting. Common choices are Latin1 or UTF8." A large "Comments" text area is at the bottom. At the bottom right, there are "Apply" and "Revert" buttons. The "Apply" button is highlighted with a red box.

スキーマの作成



The configured SQL will be displayed.
If there are no problems, select [Apply].



スキーマの作成

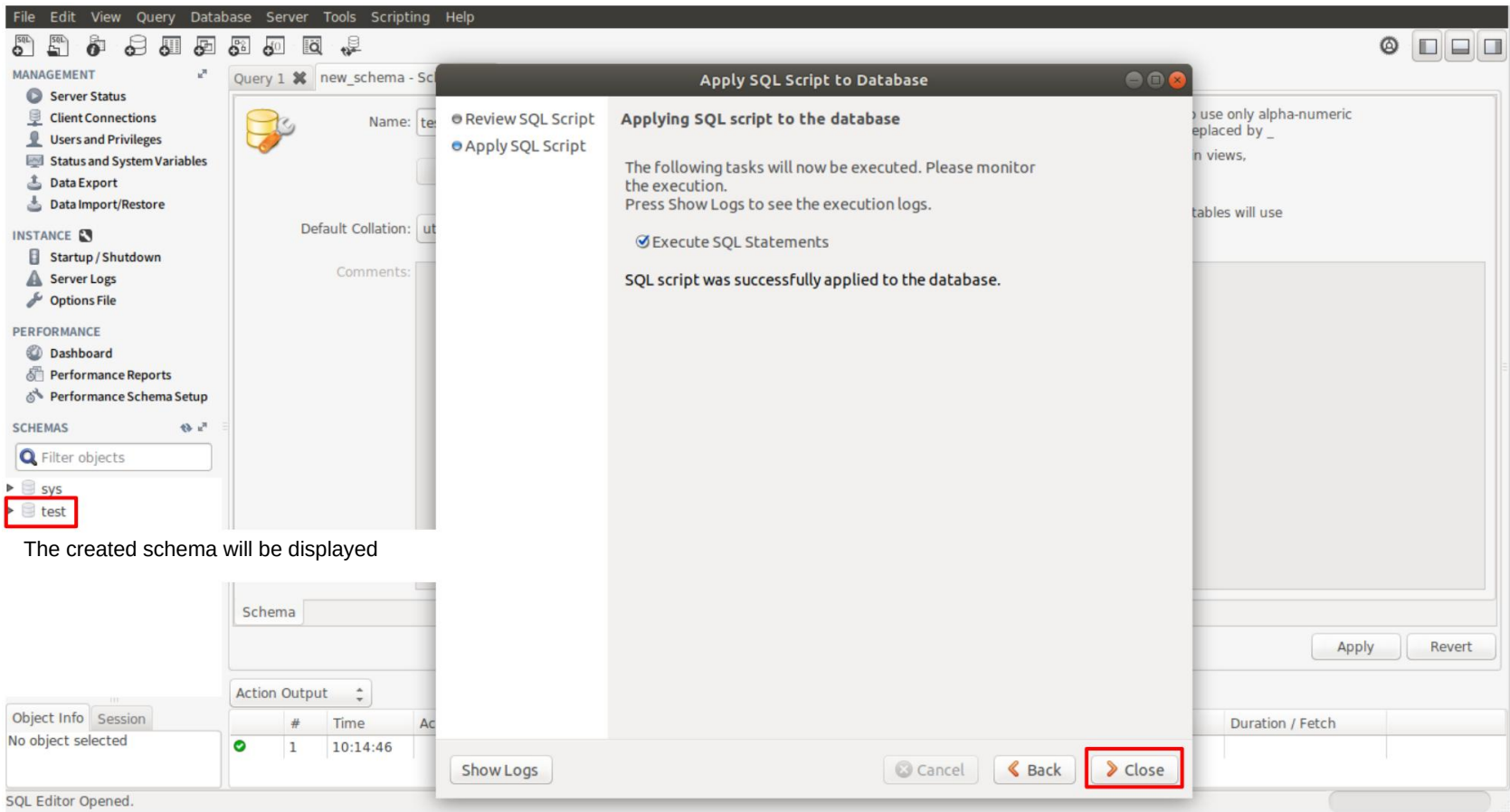
install

Create table

Creating a user

If there are no problems with the settings, you will be redirected to the creation completion screen.

Additionally, the created schema will be added to the schema list on the left side of the MySQL Workbench screen.



The created schema will be displayed

テーブルの作成

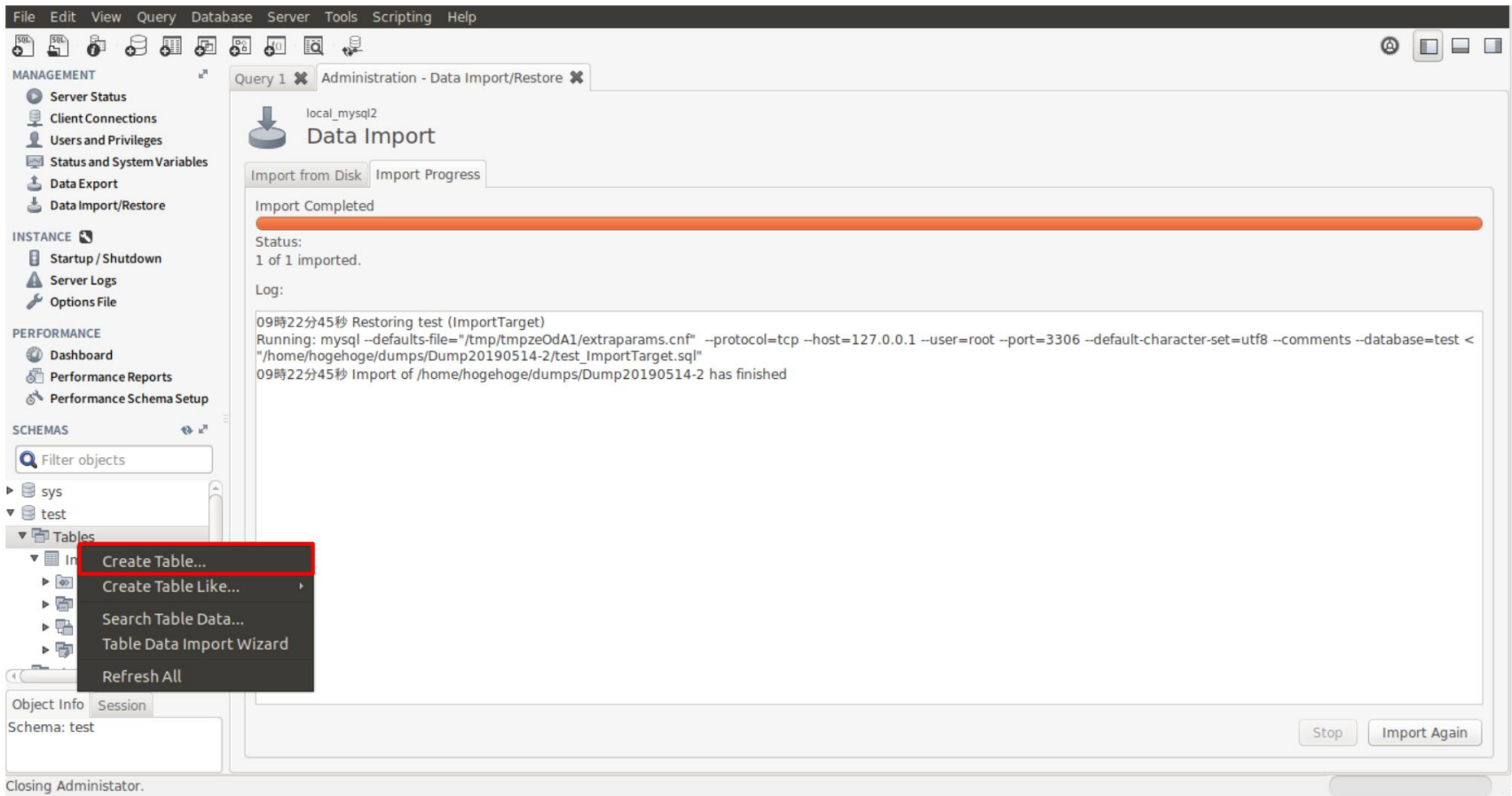
install

Create table

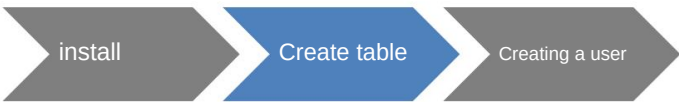
Creating a user

Next, create a table.

Expand the schema you created, right-click on Tables, and select "Create Table..."



テーブルの作成



The table creation screen will be displayed.

Set the table name (CreateTable in this case) in Name and the schema name to create the table in Schema (test in this case). Also, define the column name and data type after Column Name. After completing the input,

select [Apply].

Enter the creation table name: CreateTable - Table

Select schema to create table: test

Column Name	Datatype	PK	NN	UQ	BIN	UN	ZF	AI	G	Default / Expression
ID	INT	☒	☒	☐	☐	☐	☐	☐	☐	
Name	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	
Old	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	

Whether to use it as a primary key Whether to allow NULLs

Create columns included in the table

Column Details: Collation: *Table Default*, Comment: [text area]

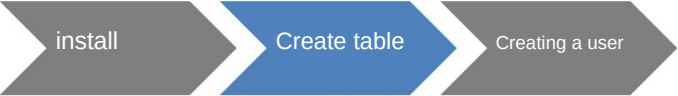
Generated Column Storage Type: ☒ Virtual ☐ Stored

Columns | Indexes | Foreign Keys | Triggers | Partitioning | Options

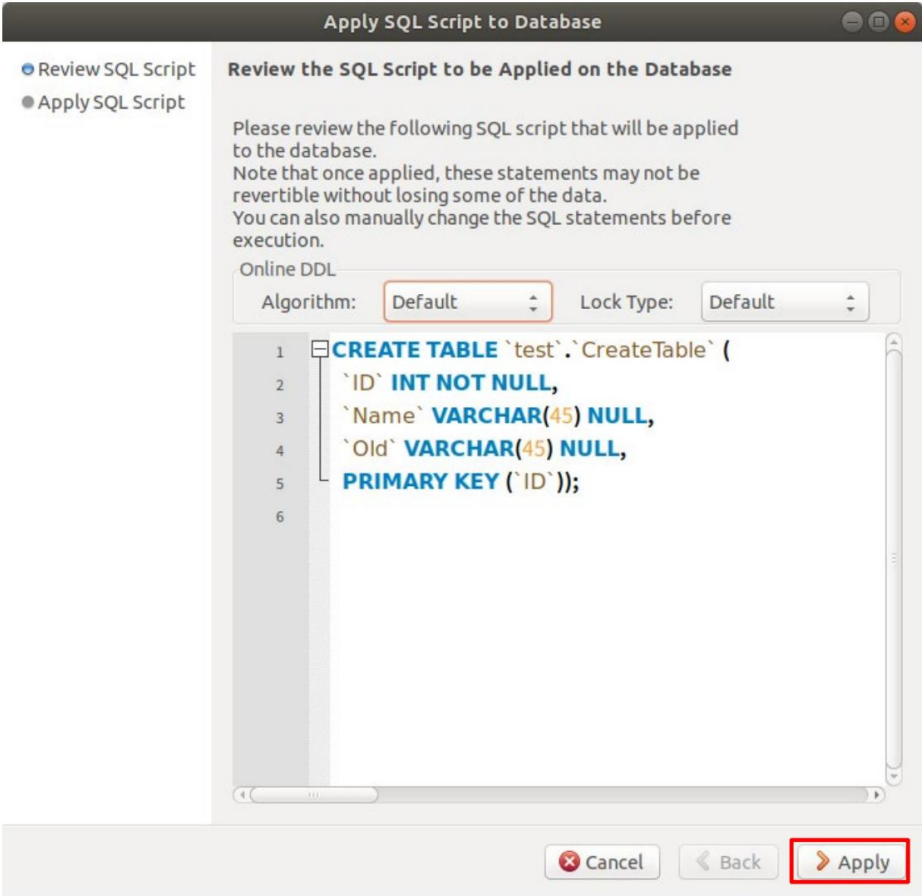
Apply Revert

•**NULL** : There is no data in the field in the RDB. You can set the column to not allow NULL. •**Primary key** : A column in which duplication of data is not allowed in an RDB. It has the role of uniquely identifying data.

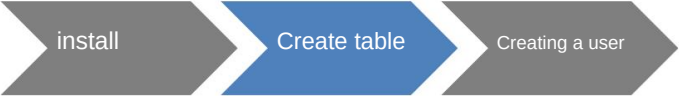
テーブルの作成



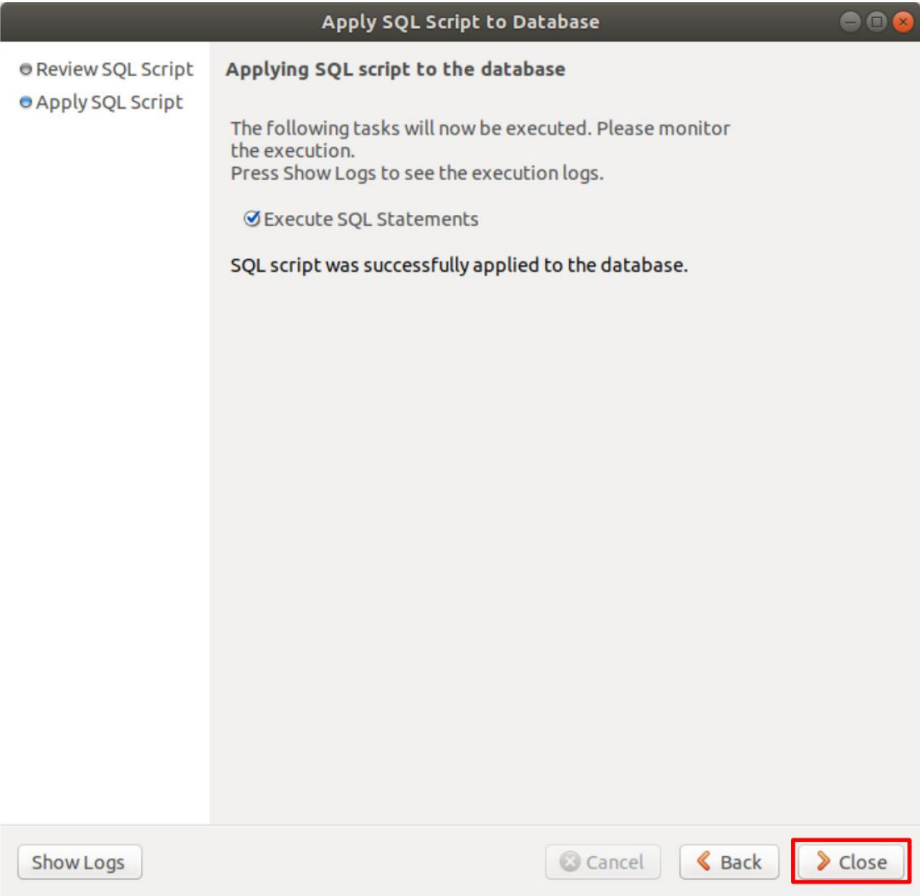
The SQL statement with the settings will be displayed.
If there are no problems, select [Apply].



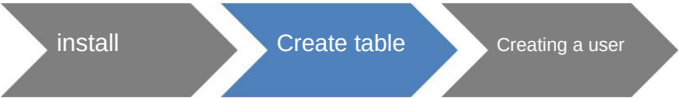
テーブルの作成



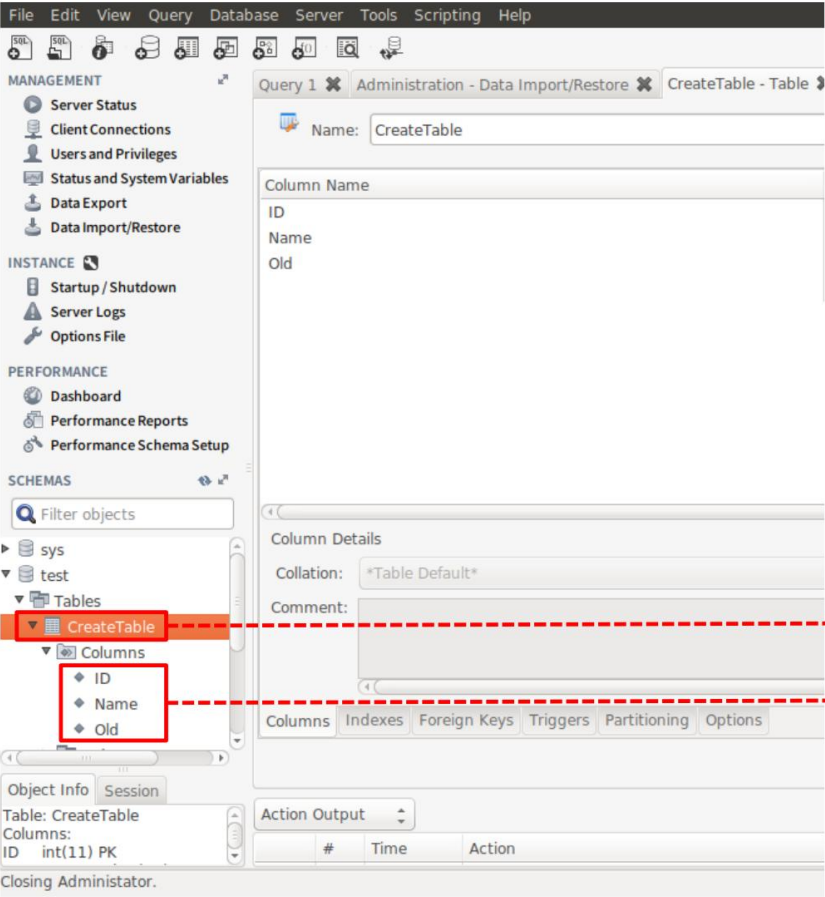
If there are no problems with the settings, you will be redirected to the creation completion screen. Select Close.



テーブルの作成



Confirm that the created table name and column name are displayed in "Schema name – Tables". The database table creation is now complete.



create table

Create column

RDBサーバのセットアップ ～KV接続用ログインユーザの作成～

install

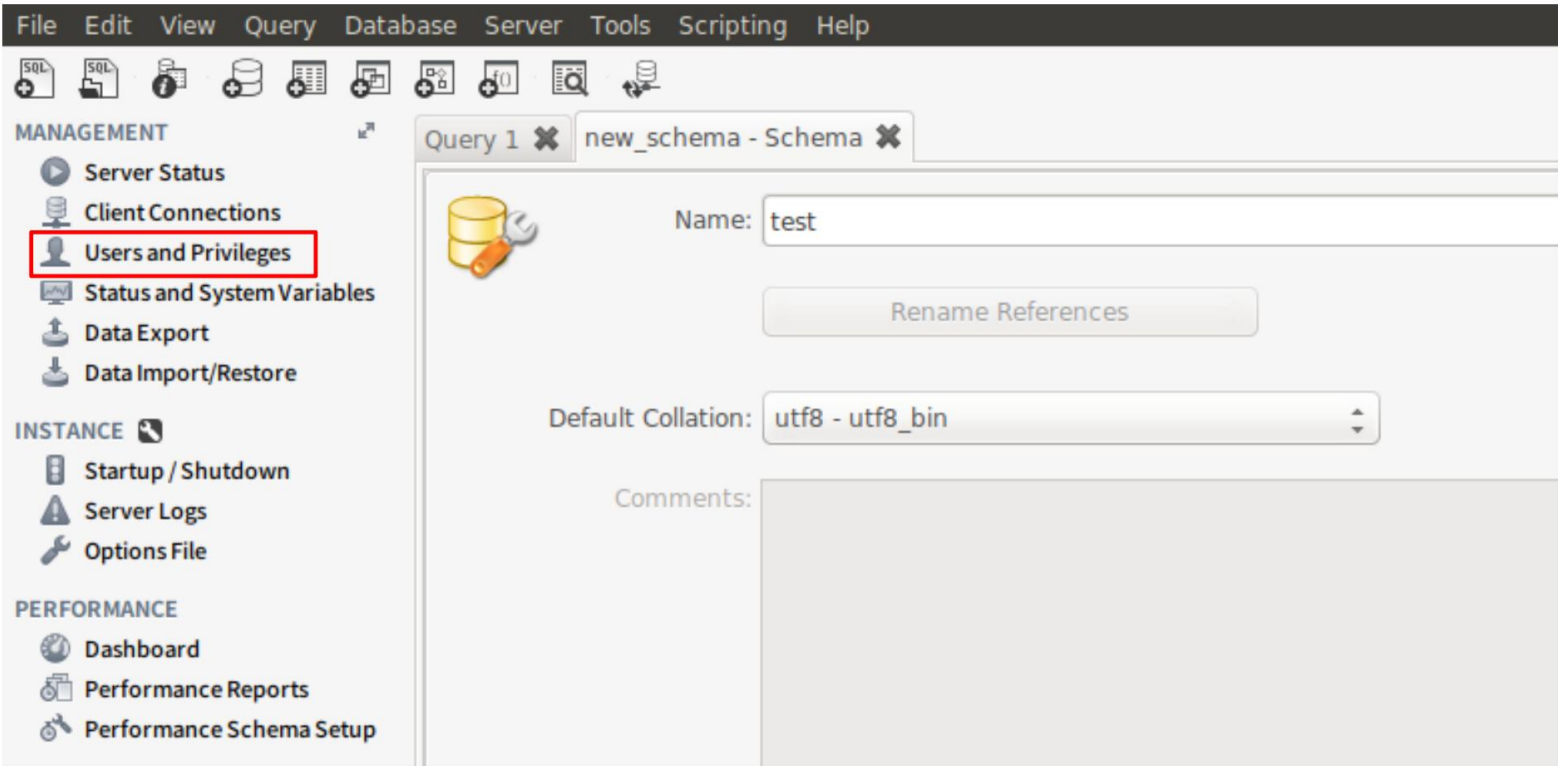
Create table

Creating a user

ログインユーザの作成

Create a new login user.

Select "Users and Privileges" on the MySQL Workbench screen.



ログインユーザの作成

install

Create table

Creating a user

The user settings screen will be displayed.
Select [Add Account] to proceed to create a new account.

Query 1 ✕ test ✕ Administration - Users and Privileges ✕

test_connect

Users and Privileges

User Accounts Select an account to edit or click [Add Account] to create a new one

User	From Host
debian-sys-maint	127.0.0.1
mysql.session	127.0.0.1
mysql.sys	127.0.0.1
root	127.0.0.1
(!) mysql.sys	localhost
(!) mysql.session	localhost

Login Account Limits Administrative Roles Schema Privileges

Login Name: You may create multiple accounts with the same name to connect from different hosts.

Authentication Type: For the standard password and/or host based authentication, select 'Standard'.

Limit to Hosts Matching: % and _ wildcards may be used

Password: Type a password to reset it.

Consider using a password with 8 or more characters with mixed case letters, numbers and punctuation marks.

Confirm Password: Enter password again to confirm.

Authentication String: Authentication plugin specific parameters.

install

Create table

Creating a user

ログインユーザの作成

Configure login user settings. These are all the settings required for KV-XCM02 to log in.

Login Name: Login user name : **"kv"** Limit

to hosts Matching: Restrict devices that the user can use by host name (IP address) : "%" (% allows all hosts)

Authentication Type: Password authentication method : "Standard" " (KV-XCM02 supports Standard.) After entering the password and confirmation password, select [Apply].

Query 1 ✕ test ✕ Administration - Users and Privileges ✕

test_connect
Users and Privileges

User Accounts

User	From Host
debian-sys-maint	127.0.0.1
mysql.session	127.0.0.1
mysql.sys	127.0.0.1
root	127.0.0.1
(f) mysql.sys	localhost
(f) mysql.session	localhost
newuser	%

Details for account newuser@%

Login Account Limits Administrative Roles Schema Privileges

Login Name: kv You may create multiple accounts with the same name to connect from different hosts.

Authentication Type: Standard For the standard password and/or host based authentication, select 'Standard'.

Limit to Hosts Matching: % % and _ wildcards may be used

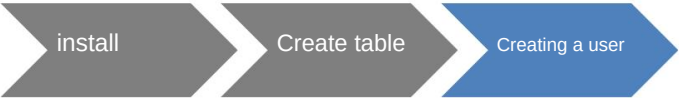
Password: Type a password to reset it.
Medium strength password.

Confirm Password: Enter password again to confirm.

Expire Password

Add Account Delete Refresh Revert Apply

ログインユーザの作成



Confirm that the newly created account (in this case, "kv") is added to the user account list.

Query 1 × test × Administration - Users and Privileges ×

test_connect

Users and Privileges

User Accounts

User	From Host
debian-sys-maint	127.0.0.1
kv	%
mysql.session	127.0.0.1
mysql.sys	127.0.0.1
root	127.0.0.1
(!) mysql.sys	localhost
(!) mysql.session	localhost

Details for account kv@%

Login

Account Limits

Administrative Roles

Schema Privileges

Login Name: kv

You may create multiple accounts with the same name to connect from different hosts.

Authentication Type: Standard

For the standard password and/or host based authentication, select 'Standard'.

Limit to Hosts Matching: %

% and _ wildcards may be used

Password:

Type a password to reset it.

Consider using a password with 8 or more characters with mixed case letters, numbers and punctuation marks.

Confirm Password:

Enter password again to confirm.

Expire Password

Add Account

Delete

Refresh

Revert

Apply

RDBサーバのセットアップ ～ログインユーザのテーブルへの権限付与～

Set access rights to the table

ユーザ権限の付与

install

Create table

Creating a user

Set the permissions for the created user.

Select the Administrative Roles tab to set administrative privileges for the selected user.

Query 1 ✕ test ✕ Administration - Users and Privileges ✕

test_connect

Users and Privileges

User Accounts

User	From Host
debian-sys-maint	127.0.0.1
kv	%
mysql.session	127.0.0.1
mysql.sys	127.0.0.1
root	127.0.0.1
(!) mysql.sys	localhost
(!) mysql.session	localhost

Details for account kv@%

Login Account Limits Administrative Roles Schema Privileges

Role	Description
☒ DBA	grants the rights to perform all tasks
☐ MaintenanceAdmin	grants rights needed to maintain server
☐ ProcessAdmin	rights needed to assess, monitor, and kill any user process running in server
☐ UserAdmin	grants rights to create users logins and reset passwords
☐ SecurityAdmin	rights to manage logins and grant and revoke server and database level perm
☐ MonitorAdmin	minimum set of rights needed to monitor server
☐ DBManager	grants full rights on all databases
☐ DBDesigner	rights to create and reverse engineer any database schema
☐ ReplicationAdmin	rights needed to setup and manage replication
☐ BackupAdmin	minimal rights needed to backup any database

Global Privileges

- ☐ ALTER
- ☐ ALTER ROUTINE
- ☐ CREATE
- ☐ CREATE ROUTINE
- ☐ CREATE TABLESPACE
- ☐ CREATE TEMPORARY TABLES
- ☐ CREATE USER
- ☐ CREATE VIEW
- ☐ DELETE
- ☐ DROP
- ☐ EVENT
- ☐ EXECUTE
- ☐ FILE
- ☐ GRANT OPTION
- ☐ INDEX
- ☐ INSERT
- ☐ LOCK TABLES
- ☐ PROCESS

Revoke All Privileges

Add Account Delete Refresh Revert Apply

install

Create table

Creating a user

ユーザ権限の付与

By selecting the "Schema Privileges" tab, you can select privileges for each schema for the selected user.
Select Add Entry.

Query 1 ✕ test ✕ Administration - Users and Privileges ✕

test_connect

Users and Privileges

Click here to display a list of schemas.

User Accounts

User	From Host
debian-sys-maint	127.0.0.1
kv	%
mysql.session	127.0.0.1
mysql.sys	127.0.0.1
root	127.0.0.1
(!) mysql.sys	localhost
(!) mysql.session	localhost

Add Account Delete Refresh

Details for account kv@%

LoginAccount LimitsAdministrative RolesSchema Privileges

SchemaPrivileges

Schema and Host fields may use % and _ wildcards.
The server will match specific entries before wildcarded ones.

Object Rights

☐ SELECT☐ INSERT☐ UPDATE☐ DELETE☐ EXECUTE☐ SHOW VIEW

DDL Rights

☐ CREATE☐ ALTER☐ REFERENCES☐ INDEX☐ CREATE VIEW☐ CREATE ROUTINE☐ ALTER ROUTINE☐ EVENT☐ DROP☐ TRIGGER

Other Rights

☐ GRANT OPTION☐ CREATE TEMPORARY TABLES☐ LOCK TABLES

Revoke All PrivilegesDelete EntryAdd Entry...

Unselect AllSelect "ALL"

RevertApply

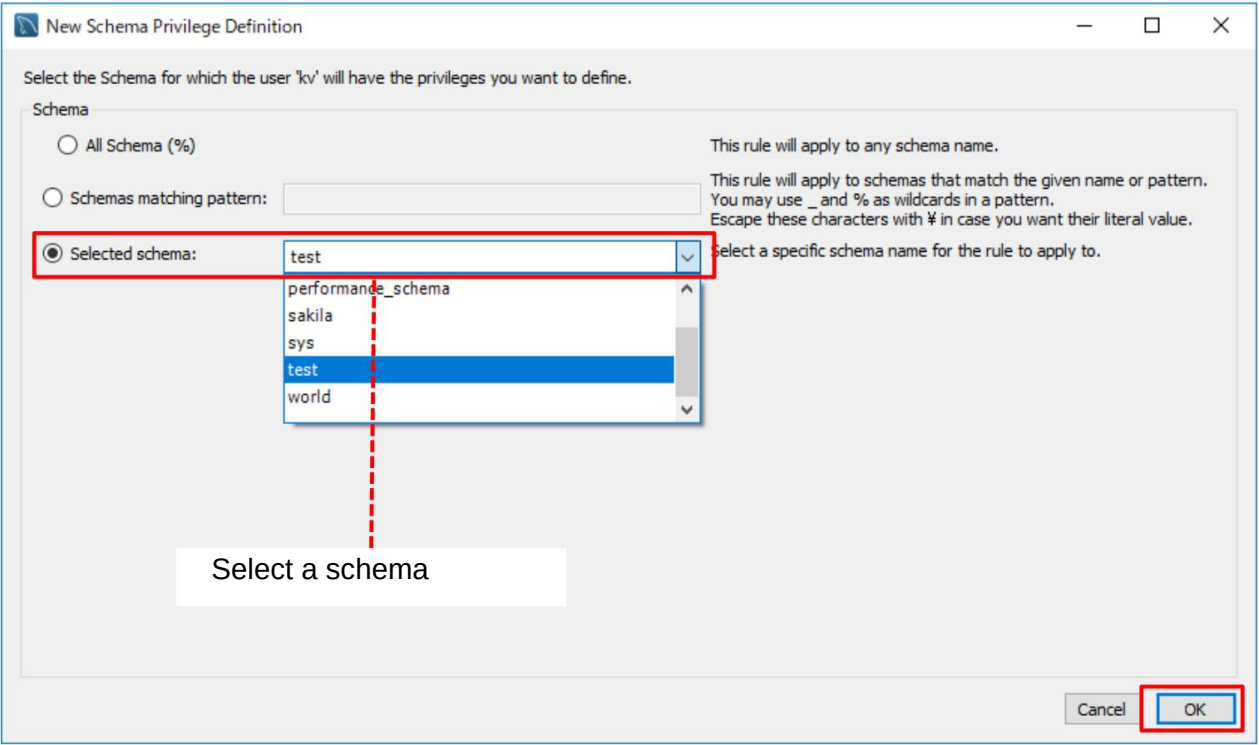
ユーザ権限の付与

install

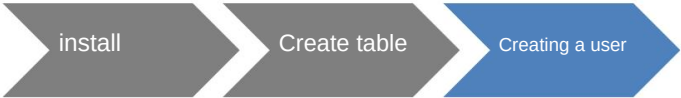
Create table

Creating a user

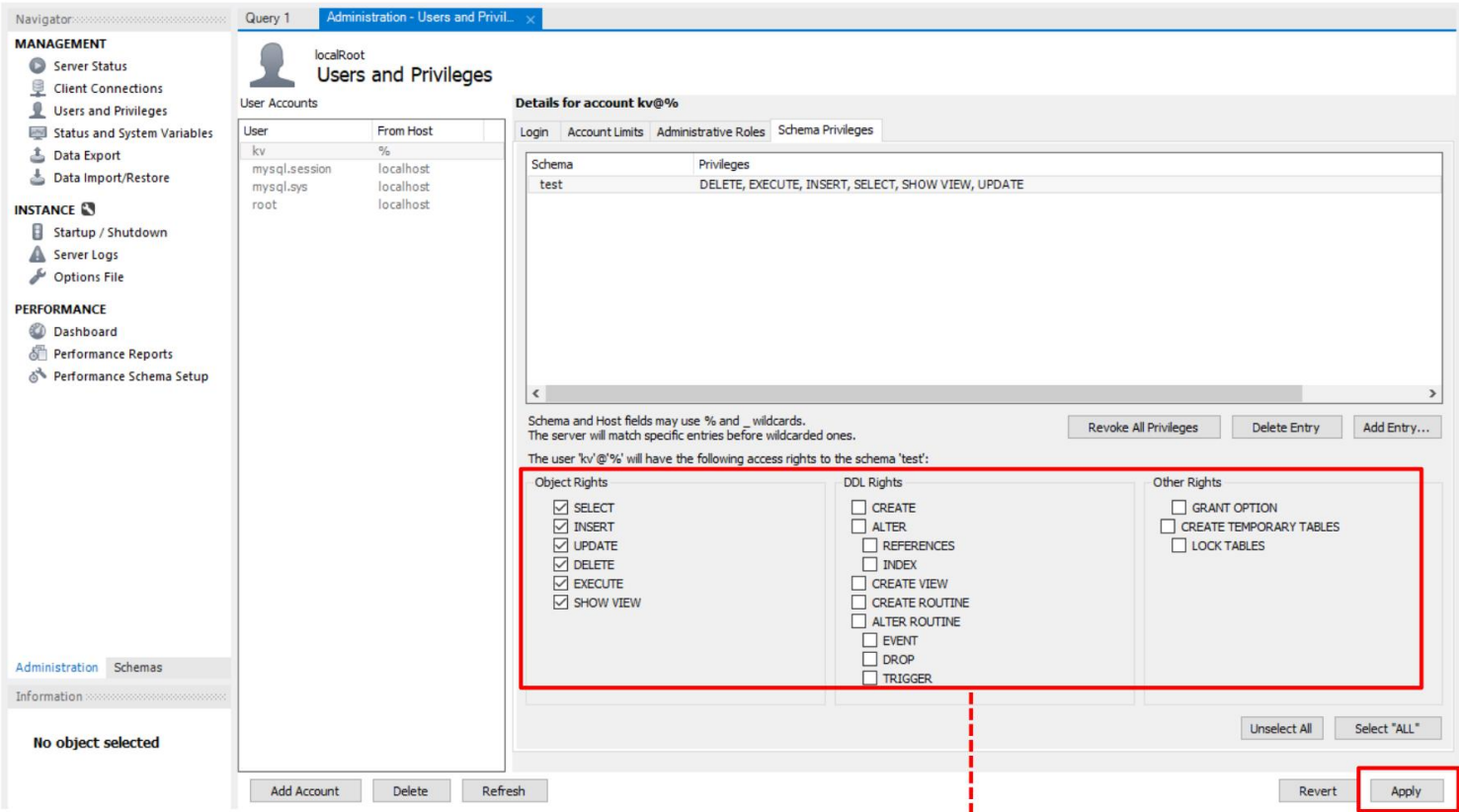
Select "Selected schema" and select the schema you want to grant permissions to from the dropdown. (This time, "test")
After making your selection, select OK.



ユーザ権限の付与



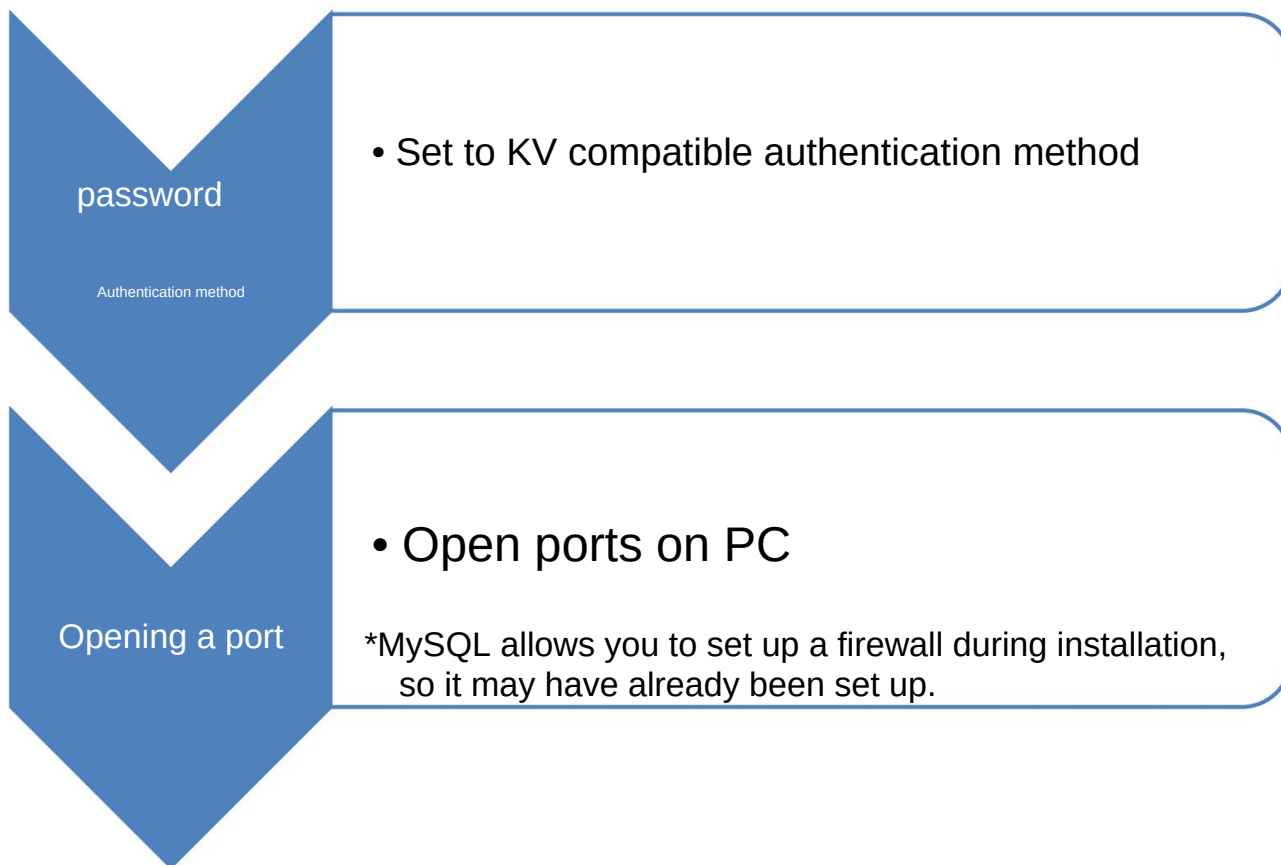
Select the required permissions and select Apply.
Add the necessary SQL privileges such as "SELECT", "INSERT", "UPDATE", and "DELETE".



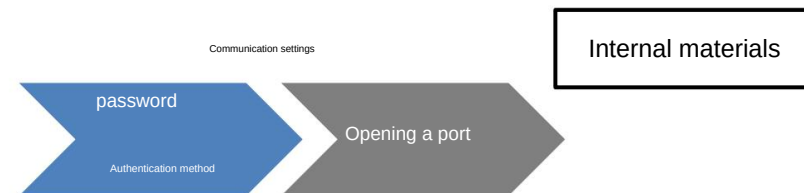
add permissions

通信設定

The following settings are required for KV (external device) to communicate with the RDB server.



通信設定 ~設定ファイル~



設定ファイルの編集

In MySQL, the password authentication method for the RDB server is configured in a dedicated configuration file.

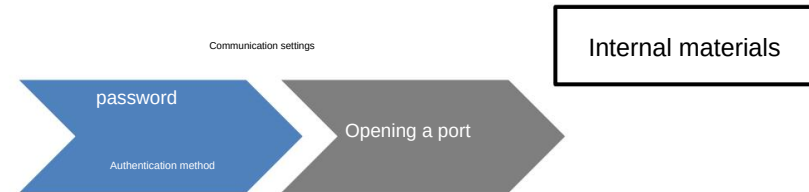
• my.ini

- Configuration file that determines the operating format of MySQL

The above files will be placed in the MySQL specified directory. `` Installation directory by default ' ' ``C:

\\ProgramData\\MySQL\\MySQL Server 8.0" in this example

*Can be edited with a text editor.



About authentication method

In MySQL 8.0 and later, the default authentication method has been changed from "Standard" to "SHA-2".

KV supports "Standard".

Setting status immediately after installing MySQL 8.0 or later

```
# The default authentication plugin to be used when connecting to the server #  
default_authentication_plugin= caching_sha2_password (*May not be listed) ÿ
```

```
default_authentication_plugin= mysql_native_password (*Add if not listed) *Be sure  
to change. (Remove the # symbol to comment out.)
```

In line with the above, the login user authentication method must also be set to **mysql_native_password** (Standard). Please refer to the following for how to change. https://www.petitmonte.com/database/mysql_authentication_plugin.html

MySQL must be restarted for the settings to take effect. Please see the appendix for instructions on how to restart.

通信設定 ~PCのポート開放~

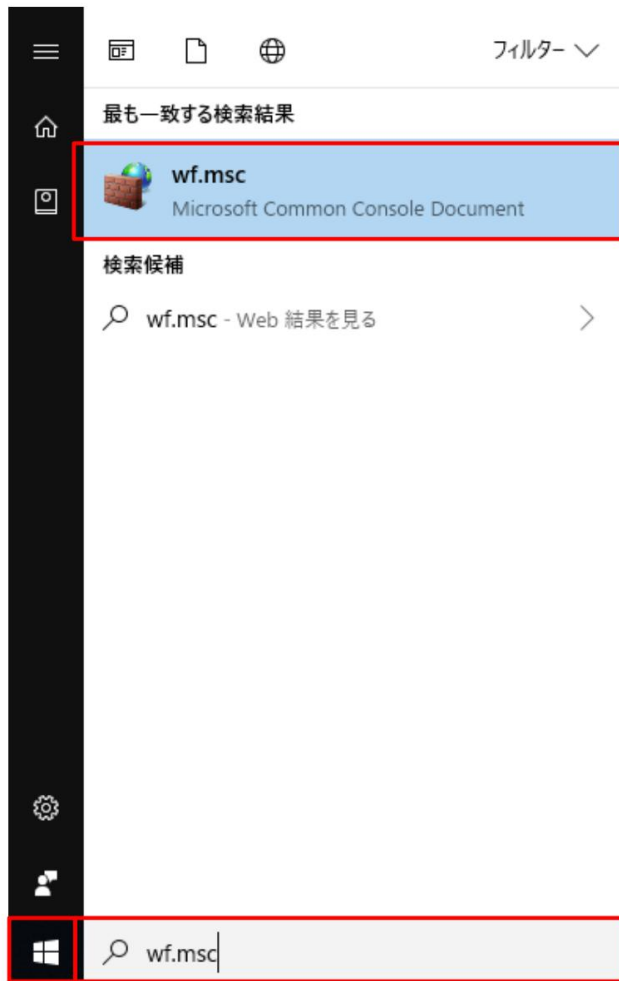
- Windows firewall settings for PC and TCP/IP connection



Windowsファイアウォールの設定

Start [wf.msc]. Click the Windows key and search for “wf.msc”

Windows Firewall Settings will start.



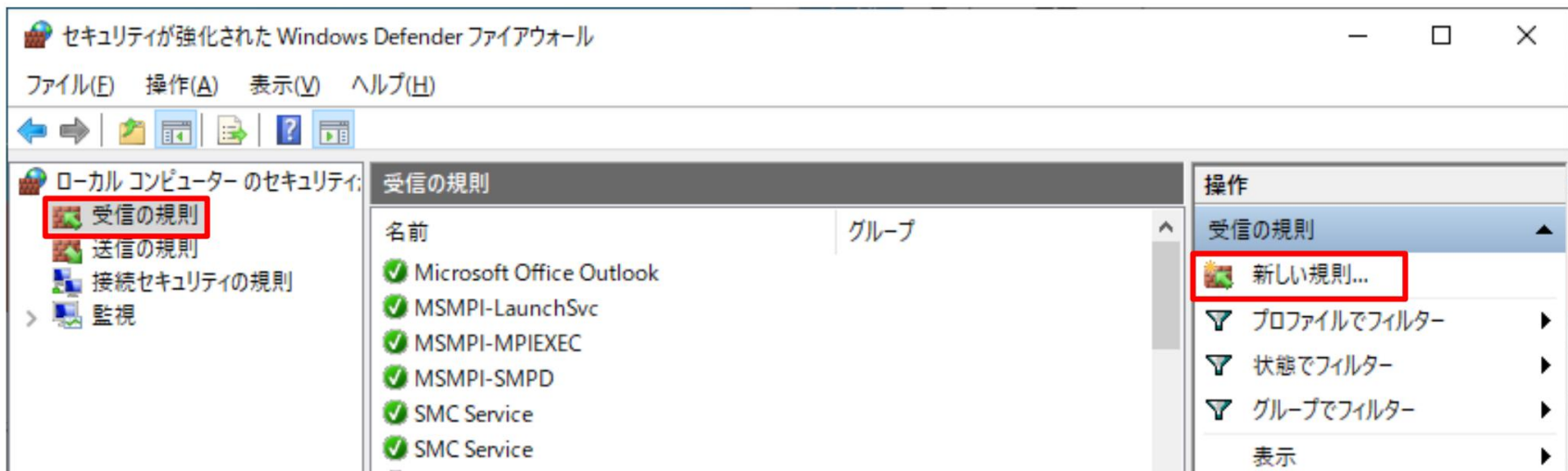
password

Opening a port

Authentication method

Windowsファイアウォールの設定

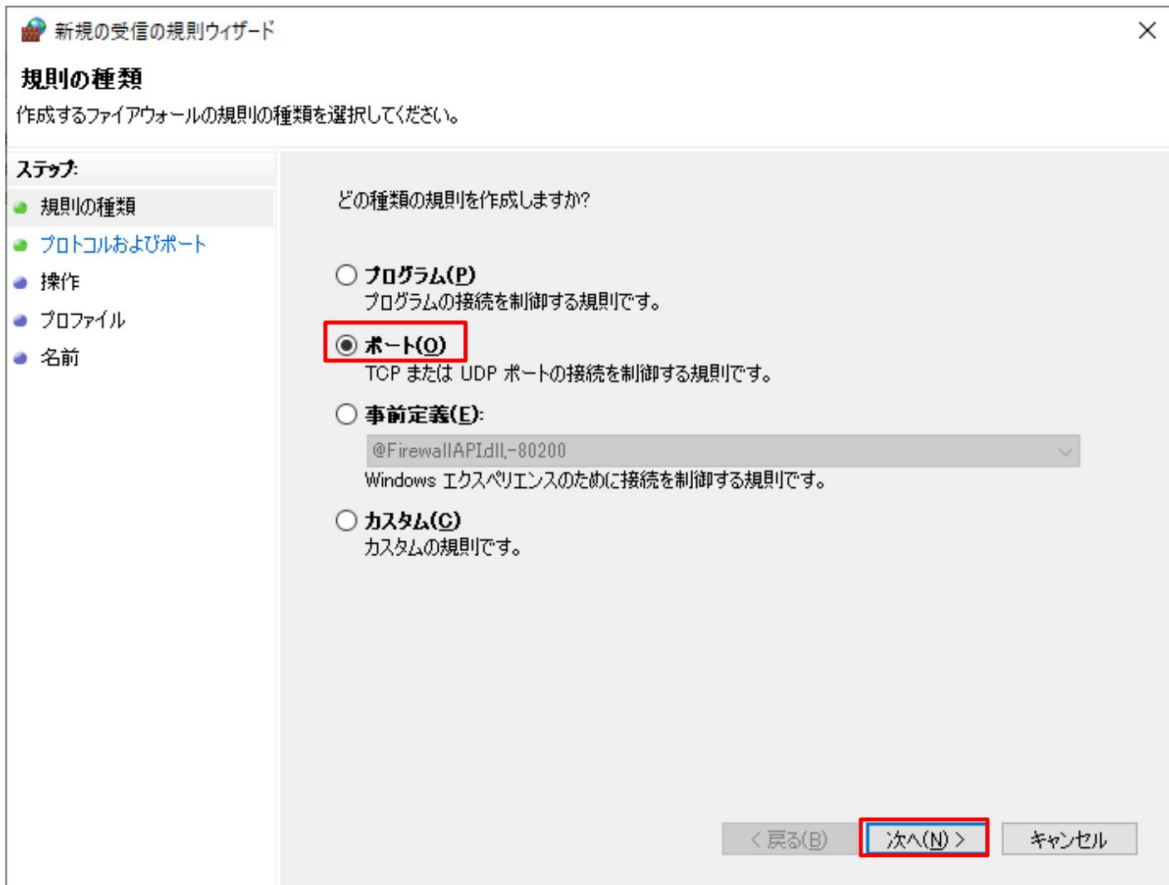
>Create a rule that allows access to TCP port 3306. Select [Reception Rules] and click [New Rule].





Windowsファイアウォールの設定

ÿ Select [Port]. ÿ Click [Next].





Windowsファイアウォールの設定

Specify TCP port number 3306. (Port number set during installation) Click [Next].

新規の受信の規則ウィザード

プロトコルおよびポート

この規則を適用するプロトコルとポートを指定してください。

ステップ:

- 規則の種類
- プロトコルおよびポート
- 操作
- プロファイル
- 名前

TCP と UDP のどちらにこの規則を適用しますか?

☒ TCP(T)

☐ UDP(U)

すべてのローカル ポートと特定のローカル ポートのどちらを対象にこの規則を適用するかを選択してください。

☐ すべてのローカル ポート(A)

☒ 特定のローカル ポート(S): 3306

例: 80、443、5000-5010

< 戻る(B) 次へ(N) > キャンセル



Windowsファイアウォールの設定

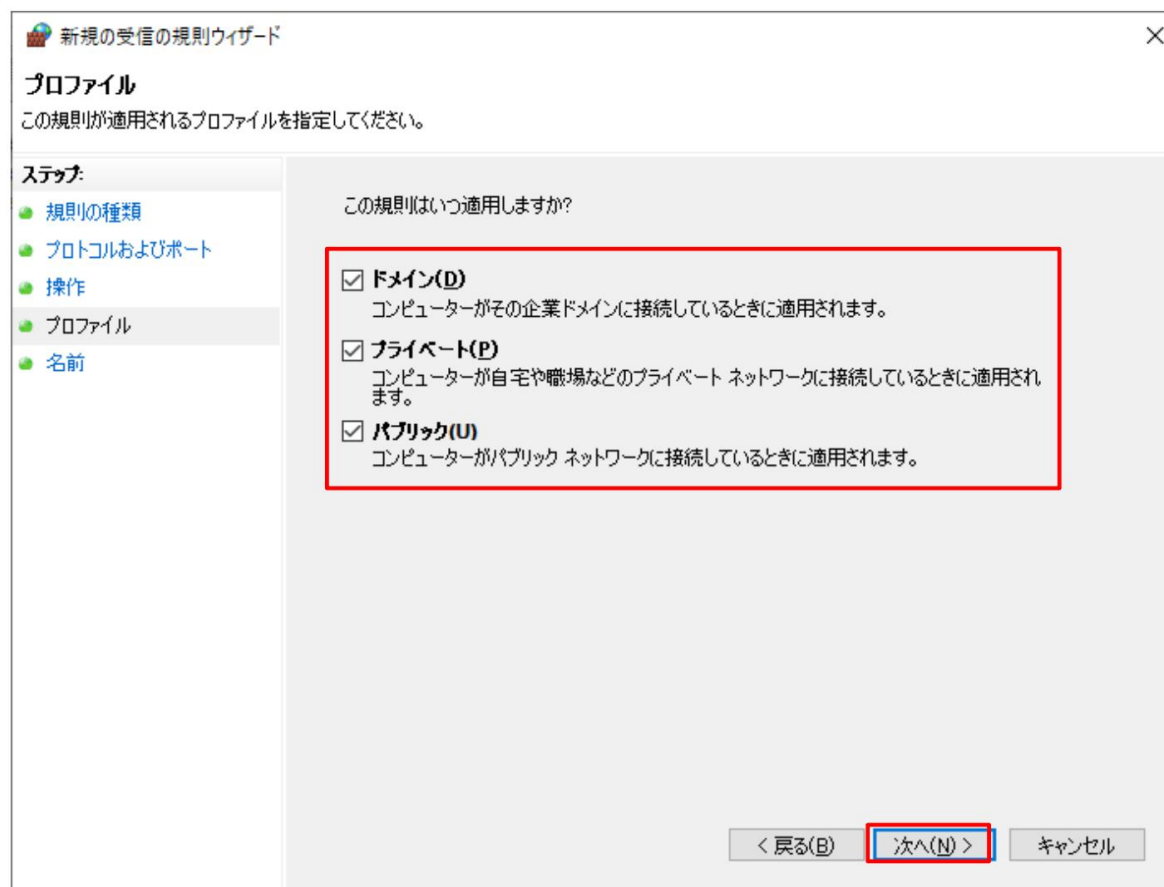
ÿ Select “Allow connection”. ÿ Click
[Next].





Windowsファイアウォールの設定

Specify the applicable profile. If you do not want to set any restrictions, check all boxes. Click [Next].





Windowsファイアウォールの設定

Set a name for the reception rule. (*The setting during installation is "Port 3306.") Click [Finish] to exit.

新規の受信の規則ウィザード

名前
この規則の名前と説明を指定してください。

ステップ:

- 規則の種類
- プロトコルおよびポート
- 操作
- プロファイル
- 名前

名前(N):
Port 3306

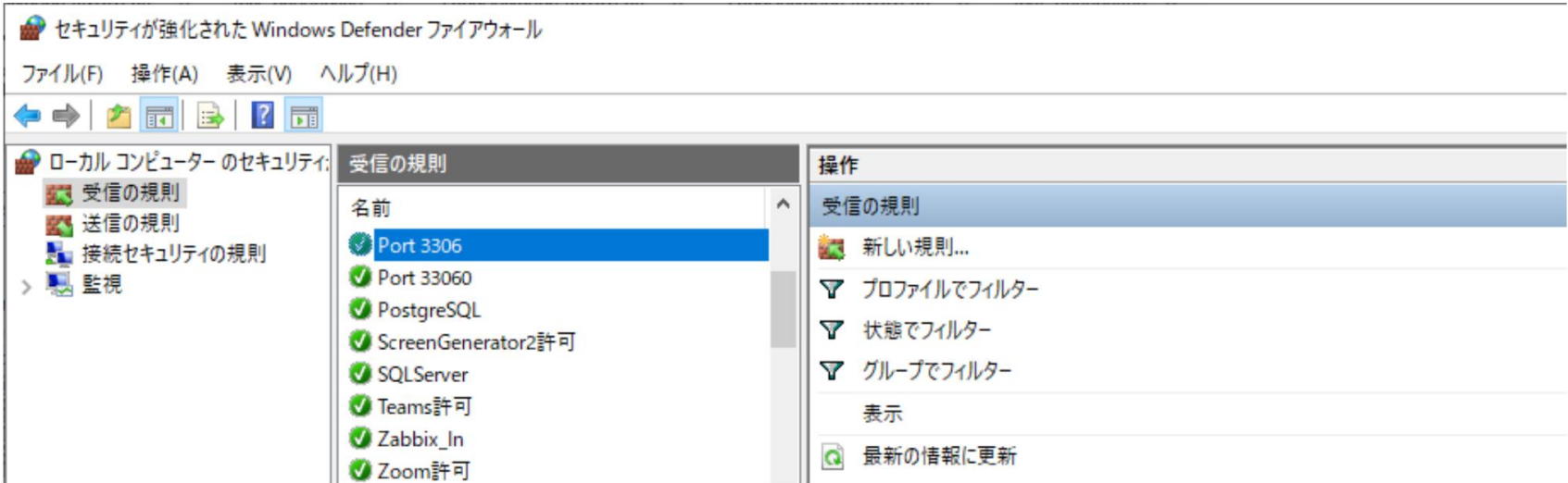
説明 (オプション)(O):

< 戻る(B) 完了(F) キャンセル



Windowsファイアウォールの設定

Port 3306 (MySQL) reception rules are applied.



付録

•How to start/restart MySQL

補足：MySQLの起動/再起動

ÿ List of Windows services



Verify that MySQL is registered as a Windows service.

To display a list of Windows services, enter something like "computer" in the search box and click "Computer Management" from the search results.



[Reference] When searching from the program list,
Windows Administrative Tools
> Computer Management

補足：MySQLの起動/再起動

The "Computer Management" screen will be displayed. Click "Services" under "Services and Applications" to display information about the service.

You can start/restart MySQL by finding the line that says "MySQL" and selecting "Start" or "Restart" from the right-click menu.

